FRP 2026 AI ENGINEERING TRACK DAY 3

How does a computer hear and speak?

Speech recognition and speech synthesis

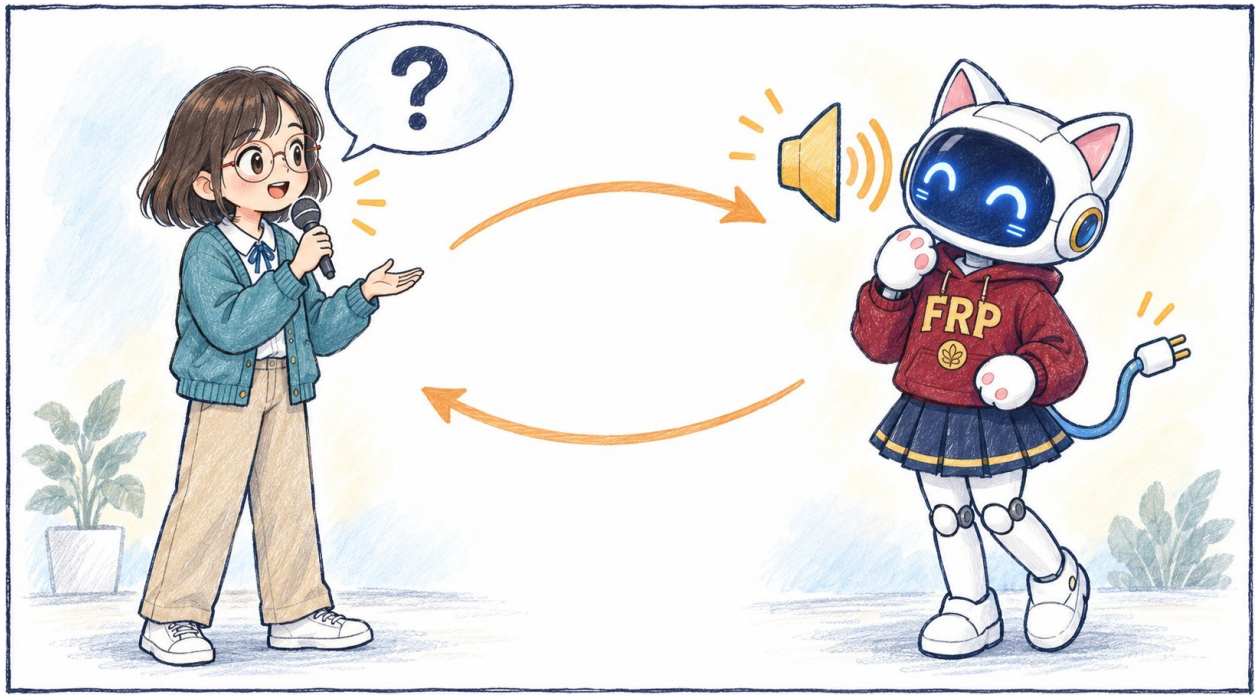
Vladimir Albrekht

ISSAI, Nazarbayev University, 26 October 2026

You speak. The computer answers, with a voice.

What does the computer hear?

Aru asks. The AI answers.



Today: how it hears, how it speaks, and how you join them.

Today: 7 parts + a workshop

A. Hear

sound becomes text

B. Inside

the ear is like an LLM

C. Mishear

mistakes, and how to check

D. Speak

text becomes sound

E. Build

hear + think + speak

F. Talk

voice coding, assistants, VAD

G. Vectors

sound as points on a map

Workshop

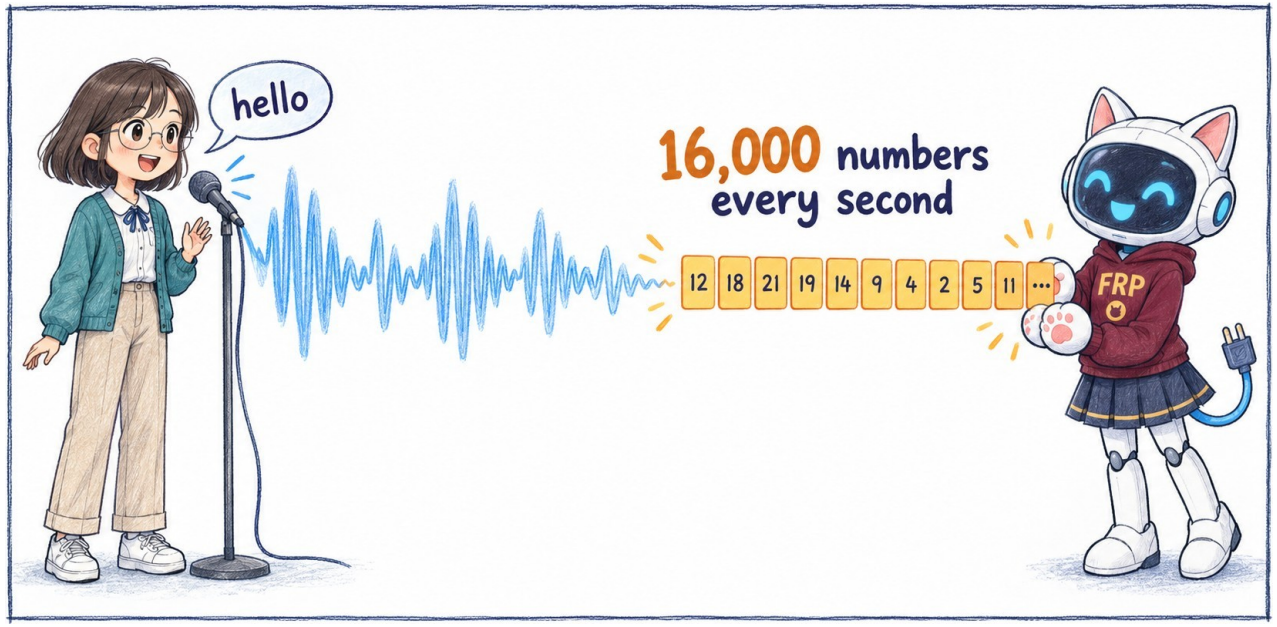
speech in your project

PART A: HOW A COMPUTER HEARS

Sound is just numbers

ASR = automatic speech recognition: sound in, text out.

Sound = a long list of numbers

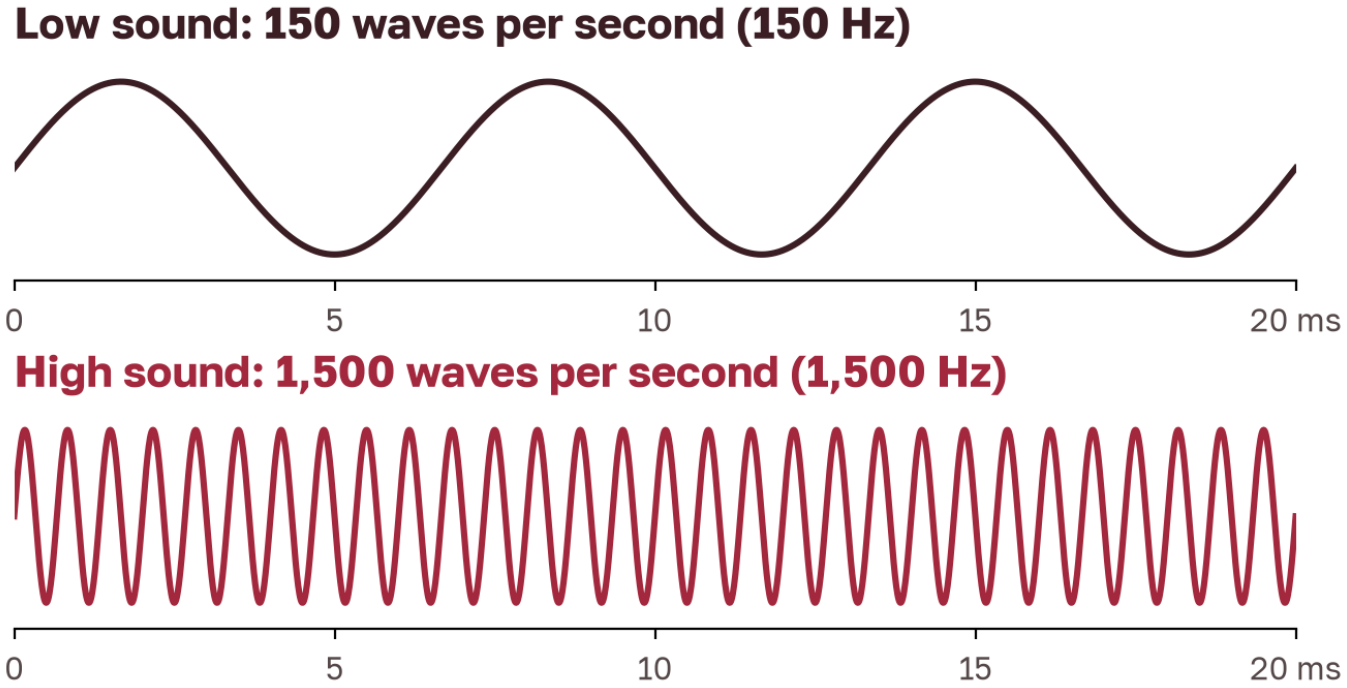


16,000

numbers every second of your voice

The computer does not hear words. It gets numbers.

Low sound, high sound



20-20,000 Hz

what a young ear hears

90-255 Hz

a speaking voice

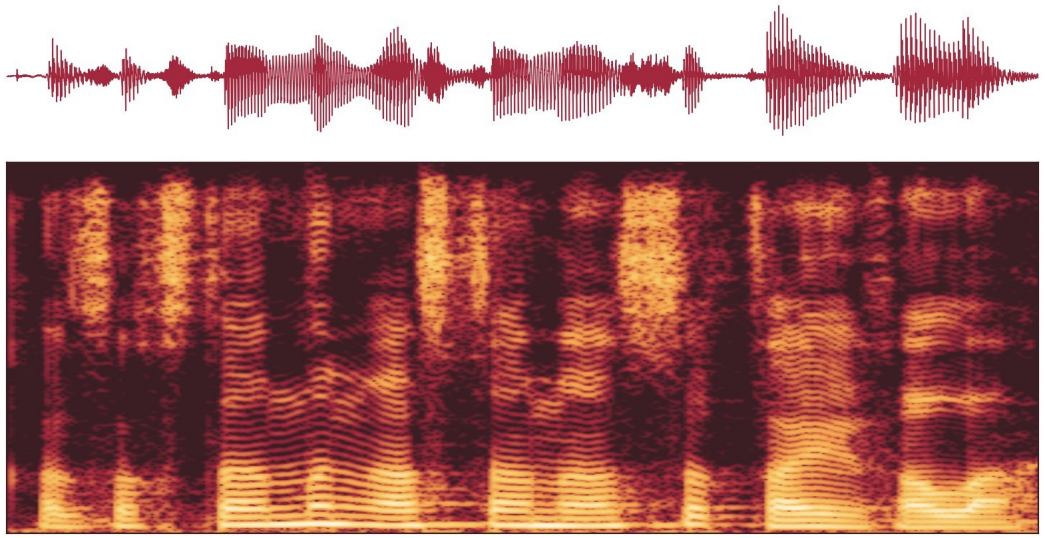
With age: **high sounds** go first.

NEW WORD

hertz (Hz) = waves per second

Low sound = slow wave. High sound = fast wave.

A spectrogram: a picture of sound



→ time

↑ low to high sounds

bright = loud

top: the wave bottom: the spectrogram

“Қазақстанның астанасы қай қала?”

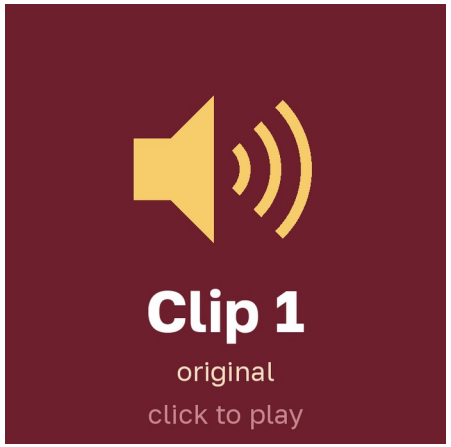
NEW WORD

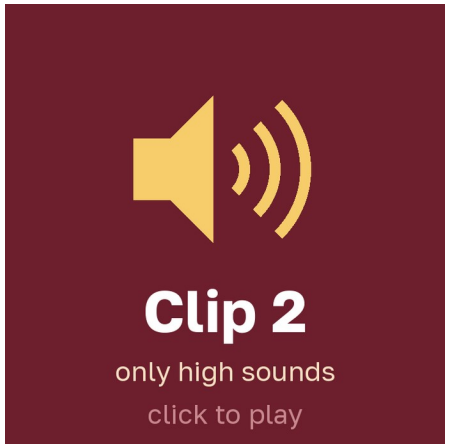
spectrogram = a picture of sound over time

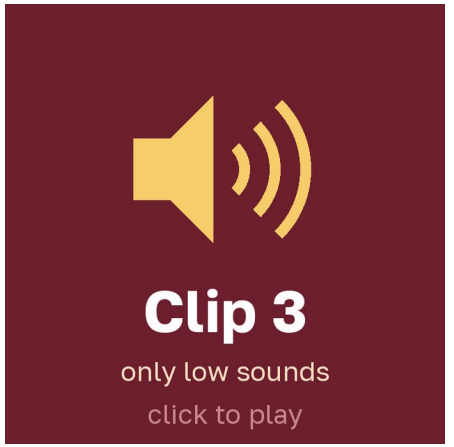
The model reads this picture, like you read a page.

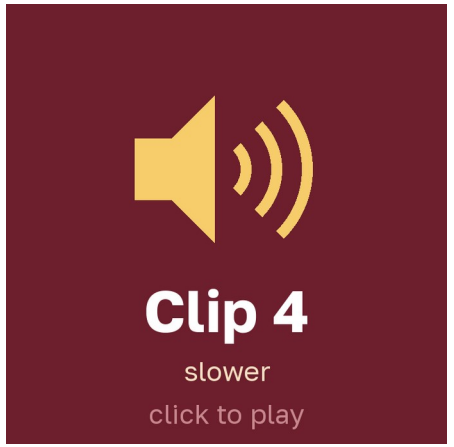
ACTIVITY, 4 MINUTES

Yanny or Laurel?





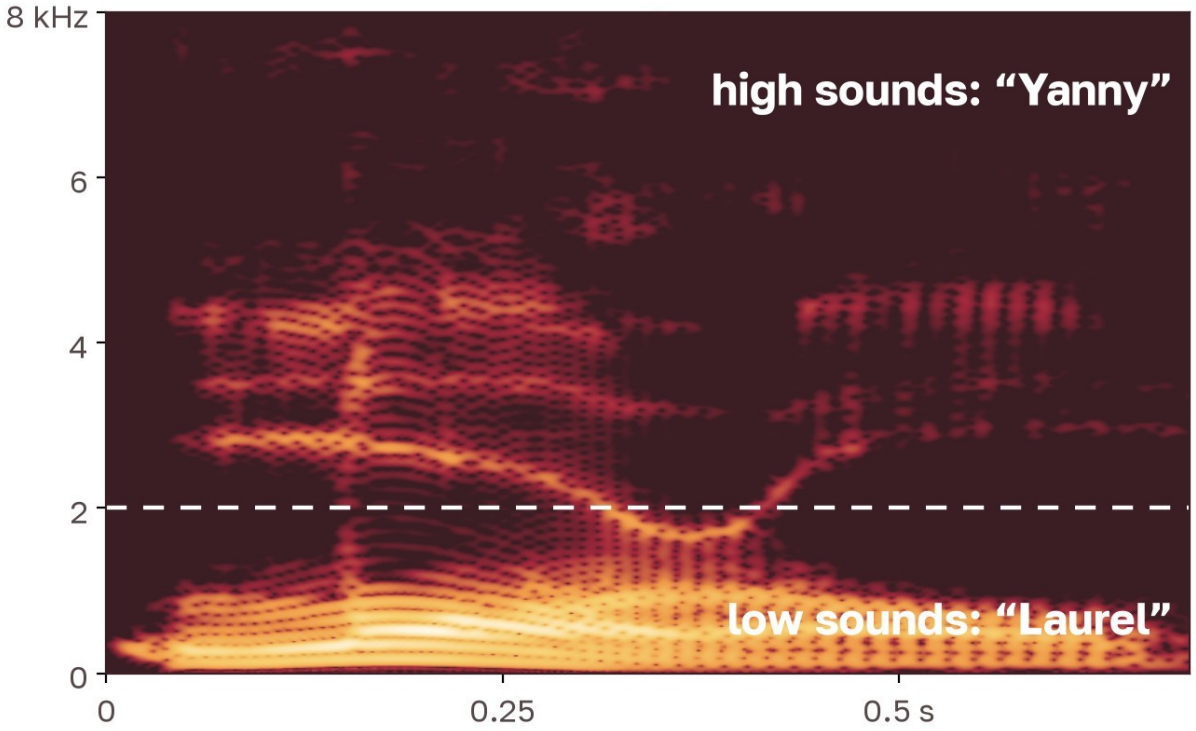




Yanny ___	Yanny ___	Yanny ___	Yanny ___
Laurel ___	Laurel ___	Laurel ___	Laurel ___

One recording. Hands up: what do you hear?

Why? Two words in one sound



2007: a singer records “laurel” for a dictionary website

2018: a student posts it, and the world argues

53 % Laurel **47 %** Yanny

Older ear = fewer high sounds = Laurel

Small speaker = fewer low sounds = Yanny

Twitter poll, more than 500,000 people

Your ear and your speaker choose the word.

What do our AI ears hear?

Clip	Whisper	GigaAM
1 original	**Laurel**	**laural**
2 only high sounds	Ewing.	mary
3 only low sounds	Moral	noral
4 slower	**Yay!**	**yealy**

Our server, 25 Oct, the same result in 2 runs.

Change the sound a bit: a different word.

ACTIVITY, 4 MINUTES

Be the ASR

I say one sentence, fast, only once.

You write what you heard.

Which word was the hardest?

> **Rare words are hard. For you, and for the model.**

Our two ears

GigaAM

Best for Kazakh

Fast: about half a second for a sentence

Whisper

Best for Russian and English

Learned from 680,000 hours of audio

Both run on our server. You choose the ear for your language.

ACTIVITY, 4 MINUTES

Count the mistakes

Said

Ғалымдар оцелоттар жейтін жануарларды иіс арқылы, жерде қайда болғанын иіскеу арқылы тауып алады деп ойлайды.

“Scientists think ocelots find their food by smell: they sniff where it was.”

GigaAM

ғалымдар **ацелоттар** жейтін **жануарлардың иісі** арқылы жерде қайда болғанын иіскеу арқылы тауып алады деп ойлайды

3 of 15

Whisper

галымдар атсеулаттар жейтін **жануарлардың иесі** арқылы жерде қайда болғанын **иескеу** арқылы тауып алады деп ойлайды

5 of 15

NEW WORD

word error rate = wrong words ÷ all words

иесі = “its owner”: a different word!

Word error rate = wrong words ÷ all words. Lower is better.

WER, step by step

True	the	**black**	cat	sat	**on**	the	–	mat
Model	the	–	cat	sat	**in**	the	**the**	mat
		D			**S**		**I**	

S = swapped word **D** = missing word

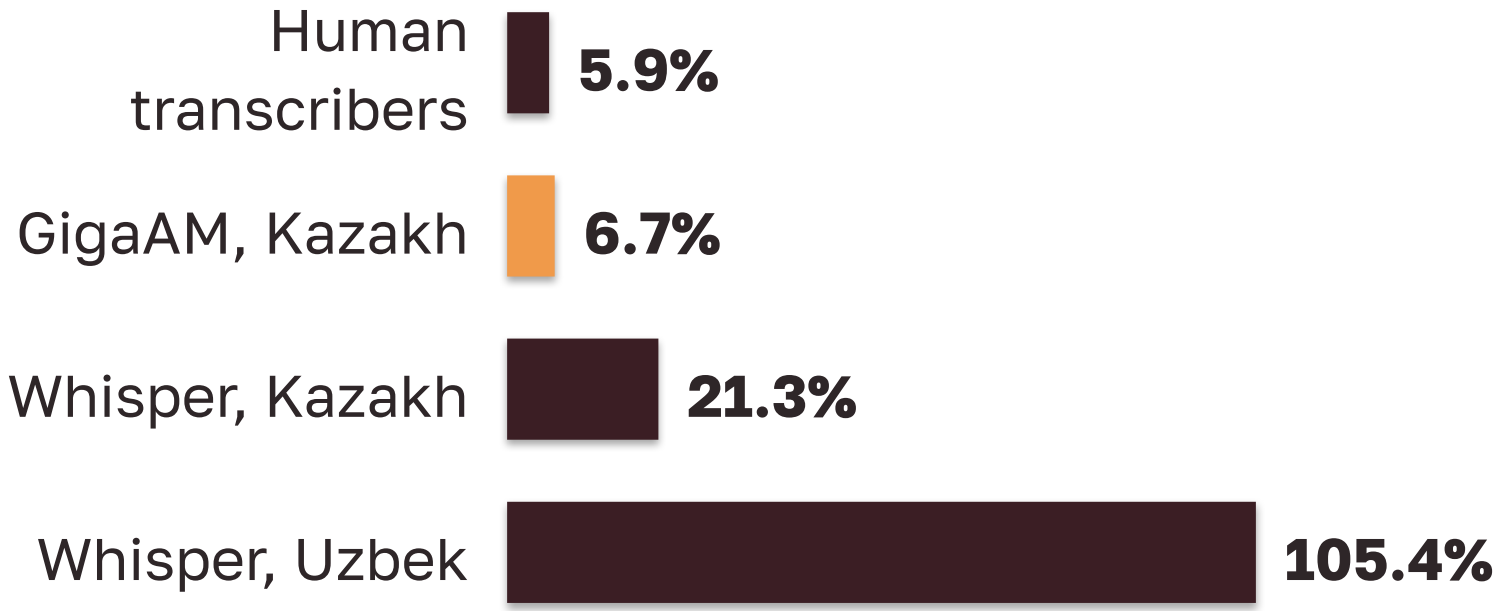
I = extra word **N** = words in the true text

WER = (S + D + I) ÷ N

= (1 + 1 + 1) ÷ 7 = **43 %**

Swapped + missing + extra words, divided by the true words.

What is a good WER?



Our rule of thumb

under 10 % good

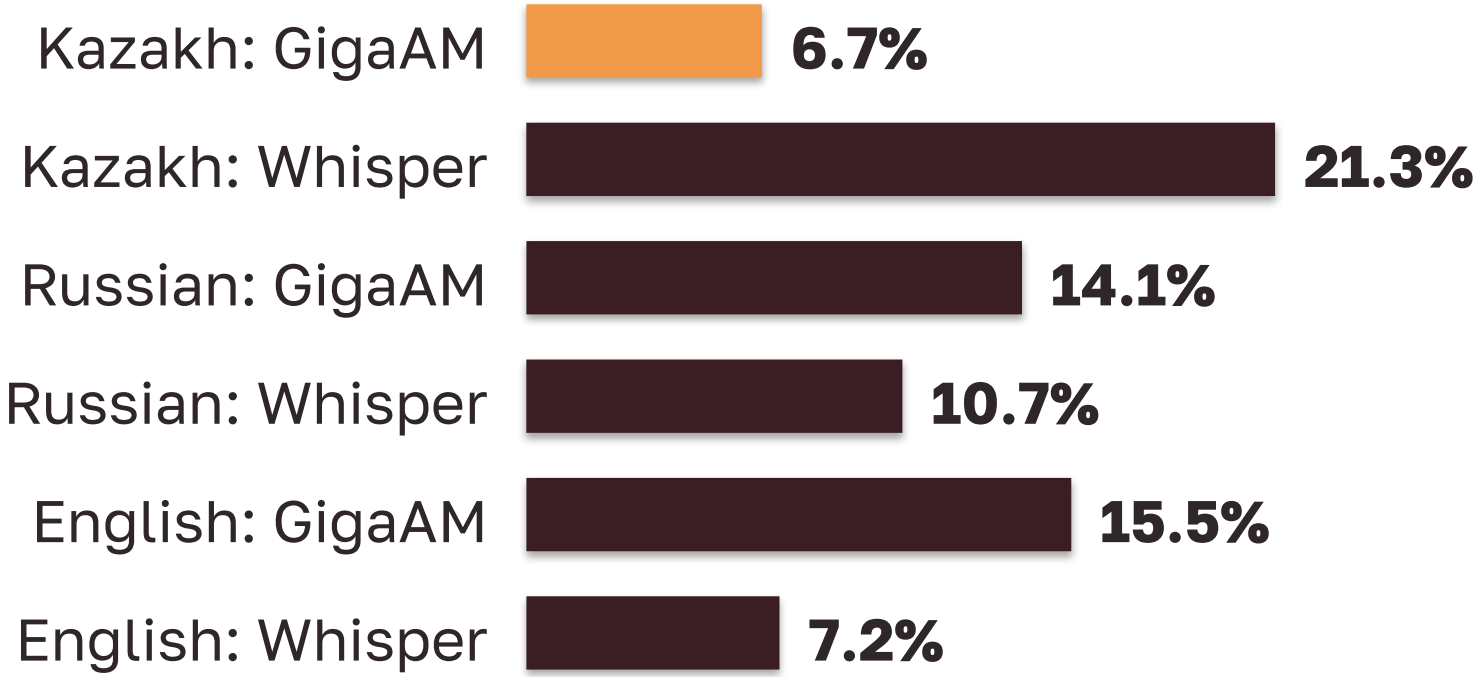
10-20 % show the text

over 30 % try another model

Humans: phone talks (Microsoft, 2016)

Lower is better. Extra words can push WER over 100 %.

Which ear for which language?



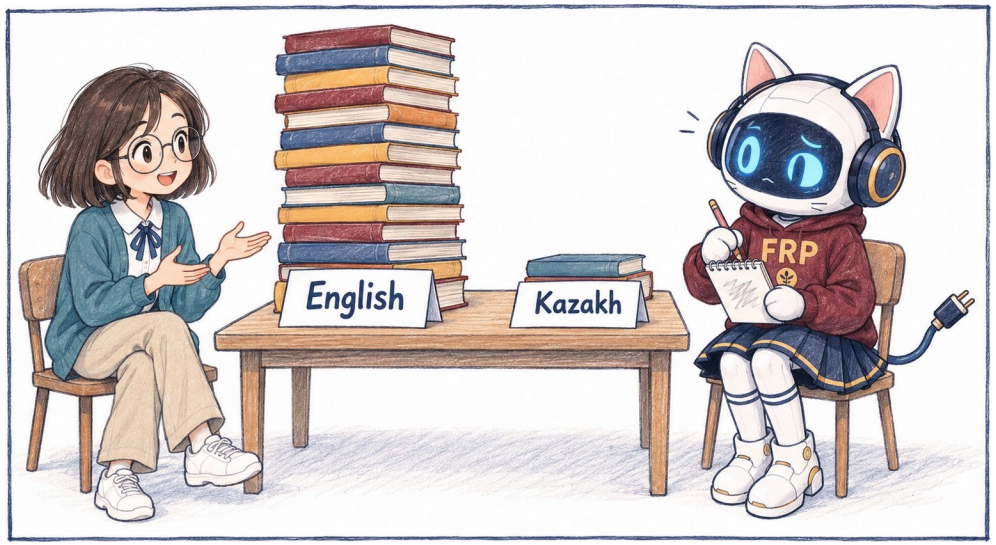
% wrong words

Our server, 3 sentences per language.

Small test: a guide, not a law.

Kazakh: GigaAM. Russian and English: Whisper.

Why does Whisper struggle with Kazakh?



Whisper's training data

438,218 h English

12 h Kazakh

Another dataset: ISSAI

1,128 h

of Kazakh in ISSAI's KSC2 corpus

NEW WORD

corpus = a big collection of recordings or texts

More Kazakh in training = better Kazakh.

Quiz time

- What does the computer get from the microphone?
- Which ear do you use for Kazakh?
- Why did it mishear “оцелоттар”?
- **Your app mishears a name. What do you do?**

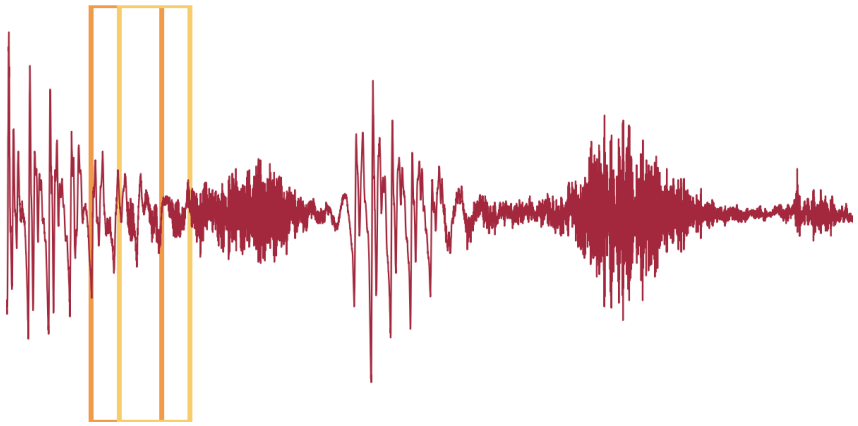
PART B: INSIDE THE EAR

How ASR works inside

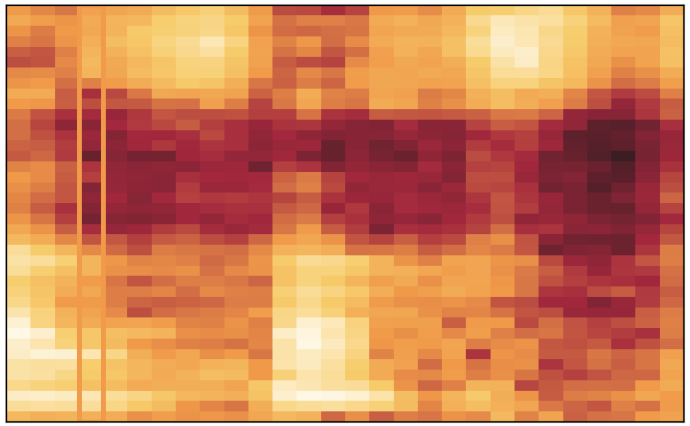
From a picture of sound to text: the same idea as the LLM yesterday.

Step 1: cut the sound into windows

1. cut the wave: 25 ms windows, every 10 ms



2. one column of numbers per window



100

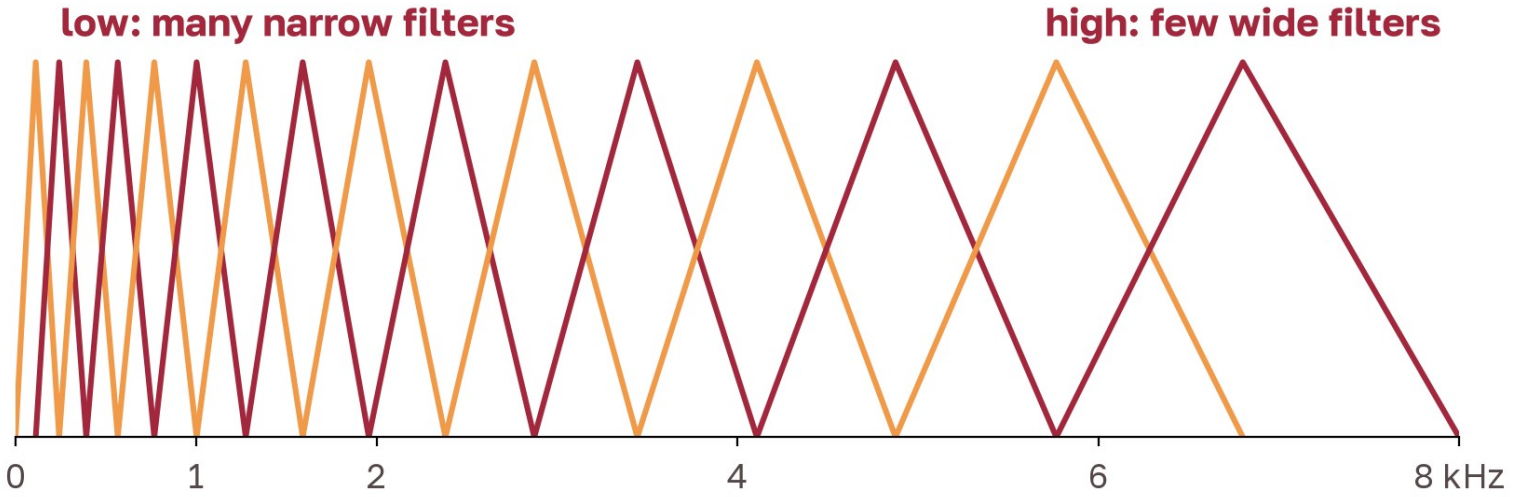
columns every second

3,000

columns in 30 s (Whisper)

Every 10 ms: one column of numbers.

Step 2: measure like a human ear



128 filters

Whisper turbo

64 filters

GigaAM

NEW WORD

mel scale = pitch steps as a human hears them

Many details in low sounds: like your ear.

Remember yesterday: the LLM

The | capital | of | Kazakhstan | is

Pieces
text → tokens

→

Numbers
a vector for each piece

→

Layers
attention: pieces look at each other

→

Next piece
“ Astana”, then repeat

It writes **one piece at a time**, and every new piece goes back in.

An LLM reads pieces and guesses the next piece.

Whisper: a short story

Sep 2022

Whisper is public

free code + model, 680,000 h

Nov 2022

ChatGPT

2 months later

Dec 2022

large-v2

and the paper

Nov 2023

large-v3

128 mel filters, 5 million h

Sep 2024

large-v3-turbo

decoder 32 → 4 layers: our ear

Open model: anyone can download it and run it, like we do on the ISSAI server.

Whisper is older than ChatGPT, and still used everywhere.

Whisper = an LLM that listens



↑ for every piece, the decoder looks at the sound

Encoder = reads. **Decoder** = writes.

NEW WORD

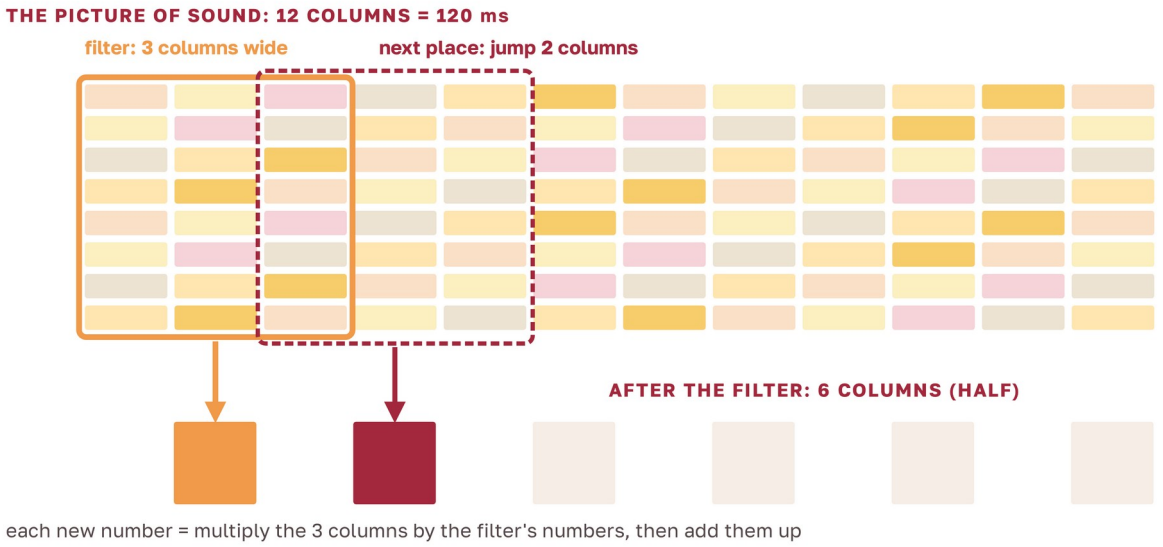
encoder = the part that reads the input

809 million

numbers in our Whisper

The encoder reads the sound. The decoder writes.

What is a conv layer (CNN)?



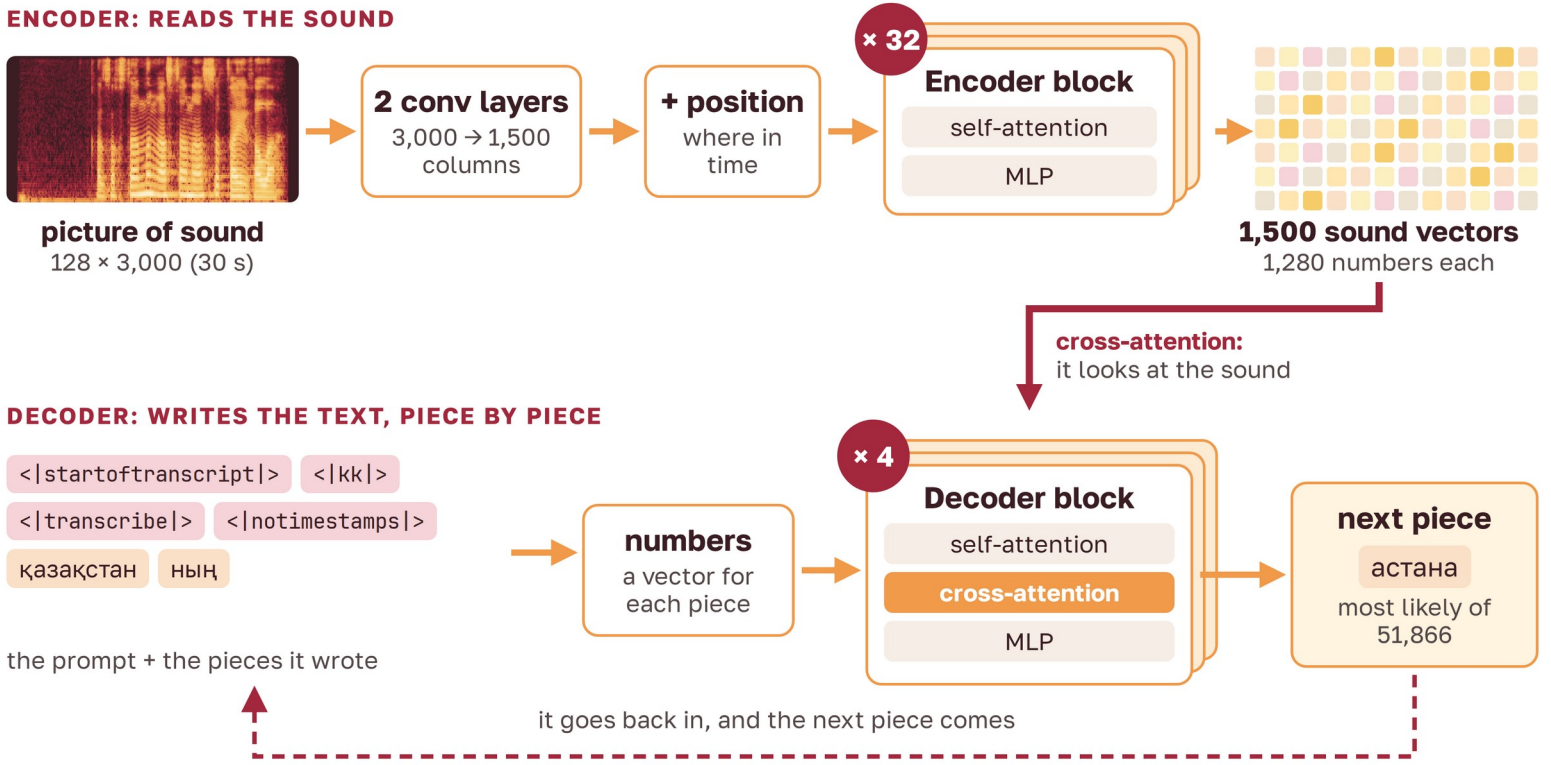
Whisper: 1,280 filters, 3 columns wide; the 2nd layer jumps 2: 3,000 → 1,500 columns

NEW WORD

CNN = sliding filters

A small filter slides over the sound and finds small patterns.

Whisper inside: the full picture



The encoder reads once. The decoder writes and looks back.

The ear has a prompt too

<|startoftranscript|> <|kk|> <|transcribe|> <|notimestamps|>

қазақстан ның астана сы қай қала <|endoftext|>

<|kk|> the language: you give it, or the model guesses it

<|transcribe|> or **<|translate|>** write what was said, or write it in English

frp-asr --lang kk = you choose the language token

Language and task are tokens: a prompt for the ear.

Another ear: one letter every 40 ms

l	l	_	a	a	u	u	_	r	r	_	e	e	l	l	_

one box = 40 ms of sound _ = blank (no new letter)

1 Join the same letters that come together: **l_au_r_el_**

2 Remove the blanks: **laurel**

NEW WORD

CTC = one letter or blank per moment, then join

GigaAM writes all letters at once.

ACTIVITY, 3 MINUTES

Be the CTC ear

1 ққ_аа_зз_аа_ққ → _____

2 мм_ее_кк_тт_ее_пп → _____

3 кк_аа_сс_сс_аа → _____

Hint for 3: look at the blank between the two “cc”.

> **Two same letters in a word need a blank between them.**

Two kinds of ears

	GigaAM (CTC)	Whisper (enc. + dec.)
Writes	letters, all at once	piece by piece, like an LLM
Knows	70 letters	51,866 pieces, 99 languages
Speed	**0.46 s**	0.89 s
Punctuation	no	yes, and it can translate
Invents text?	no	**yes: see Part C**

Whisper writes like an LLM. So it can also invent like an LLM.

How does an ear learn?

Whisper: sound + text

680,000 hours from the internet: hear the sound, write its text

GigaAM step 1

2 million hours, no text: guess the hidden sound

GigaAM step 2

50,000 hours with text: learn the letters

680,000 h

= 78 years of sound, day and night

2,000,000 h

of speech, 70+ languages

First learn the sounds, then learn the letters.

HANDS UP, 3 MINUTES

Quiz time

- How many columns of numbers in 1 second of sound?
- Why many narrow filters for low sounds?
- Which ear can invent a whole sentence? Why?
- **Be the CTC ear: “тт_оо_пп” = ?**

PART C: WHEN IT MISHEARS

Hearing is guessing

So it can be wrong, and sometimes it invents.

Rare words and names



If it never heard a word, it writes something that sounds close.

Sometimes it hears what was never said

1 in 100

Whisper transcripts got a whole sentence that nobody said (2024 study)

Most often after silence and long pauses.

About 4 in 10 of those sentences were harmful: violence, false facts.

Cut the silence = stop recording when the person stops talking.

"Careless Whisper", Koenecke et al., 2024

Cut the silence. Show the text. Let the user check.

We tried it: 1 second of hiss

The Laurel clip, only the sounds above 4 kHz: no words you can hear.

Whisper

Bye. Bye. Bye. Bye. Bye. Bye. Bye. Bye. Bye. Bye. Bye. Bye. Bye. Bye. …

178 words (above 5 kHz: “Thank you.”)

GigaAM

“e”

one letter (above 5 kHz: nothing)

No clear sound: an LLM-style ear writes what it saw most.

3 ways to check what it heard

1. Show it

Put the text on the screen

2. Ask back

“Did you say: …?”

3. Two ears

Both models agree? Probably right

Hearing is a guess. Check it, like every AI answer.

Break, 10 minutes

Next: how a computer speaks

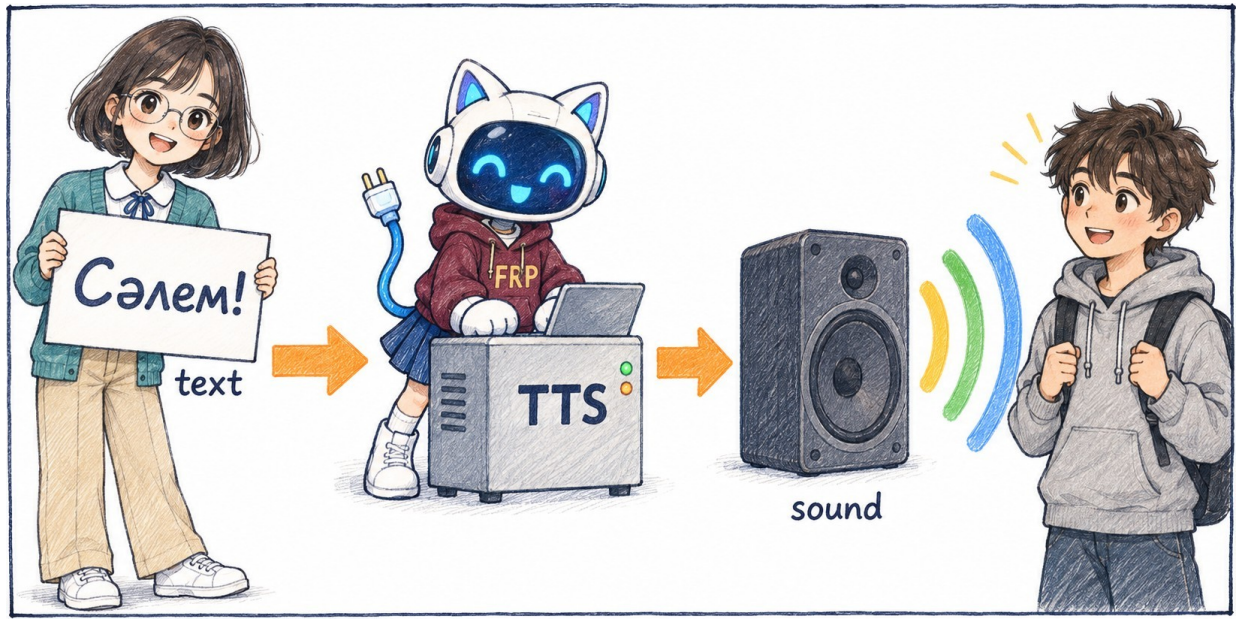
PART D: HOW A COMPUTER SPEAKS

Text becomes sound

TTS = text to speech.

LIVE

Our two voices



OmniVoice

Natural. Can copy a voice.

1.2 to 2.3 s for a sentence

Piper: class default

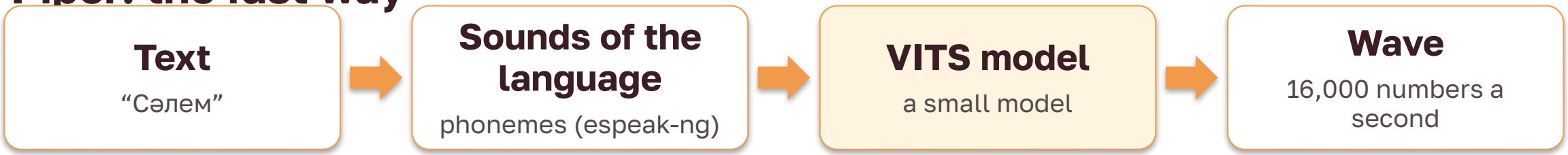
Very fast. Sounds like a robot.

0.2 to 0.7 s for a sentence

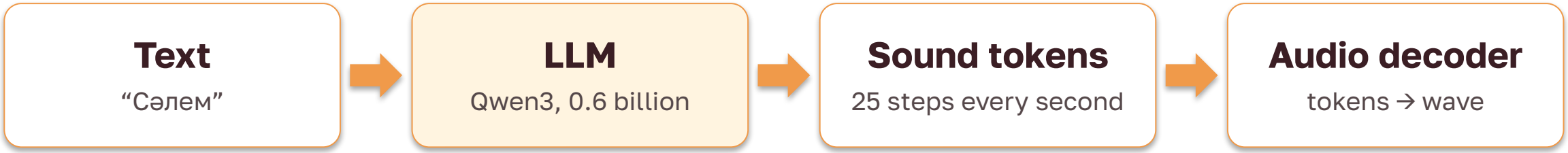
TTS is ASR backwards: text in, sound out.

How a computer speaks, inside

Piper: the fast way

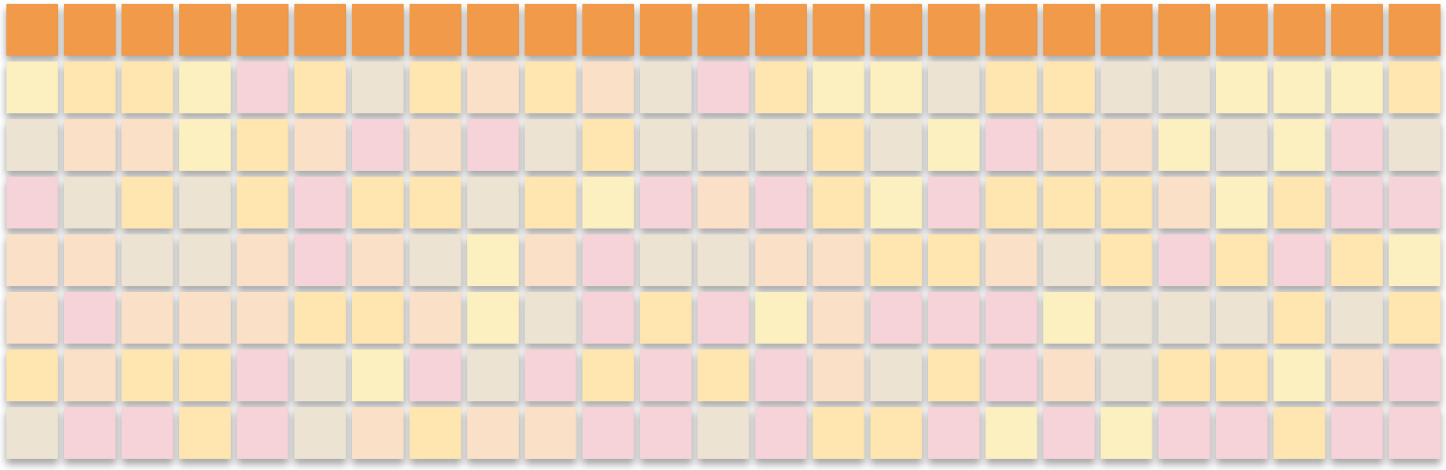


OmniVoice: an LLM that writes sound



New TTS = an LLM that writes sound tokens, not words.

Sound tokens: sound cut into pieces



← 1 second = 25 steps →

top row: the main sound lower rows: the details

25 × 8

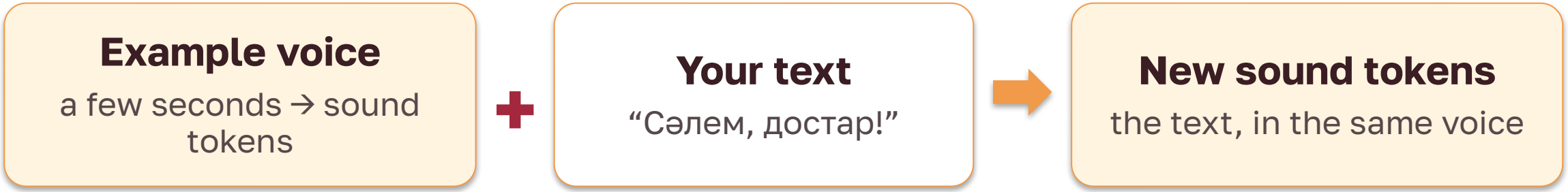
steps × tokens in 1 second

200

sound tokens in 1 second of voice

Sound can be cut into tokens. Then an LLM can write sound.

Copy a voice = a prompt

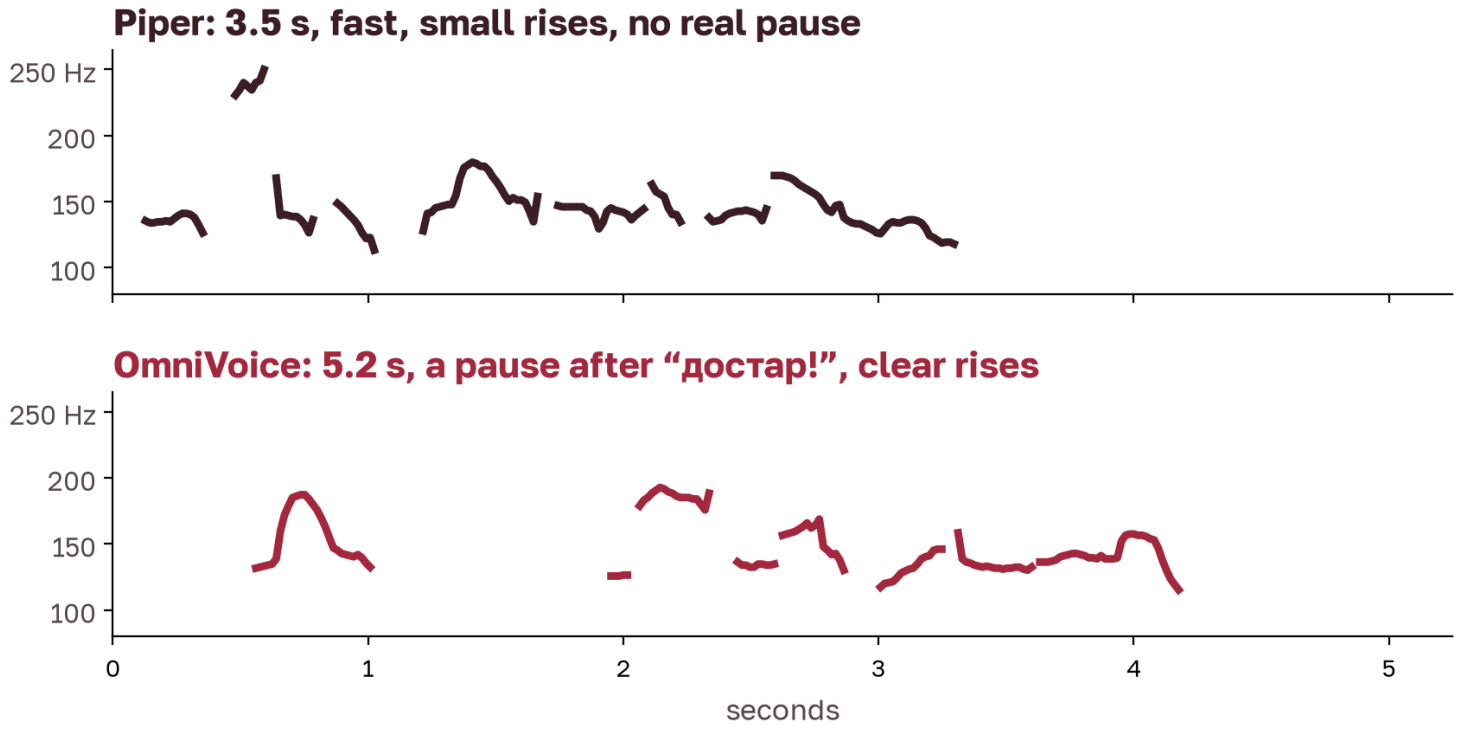


Like yesterday's prompt: what you put first changes what comes next.

No training for the new voice. A few seconds are enough.

The example voice is a prompt: same voice, new words.

What makes a voice sound nice?



Piper
fast, robot-like
click to play

OmniVoice
natural
click to play

Prosody = the melody and rhythm of speech:

pitch up and down, pauses, speed, stress

Nice speech = the right melody, pauses and stress.

LIVE

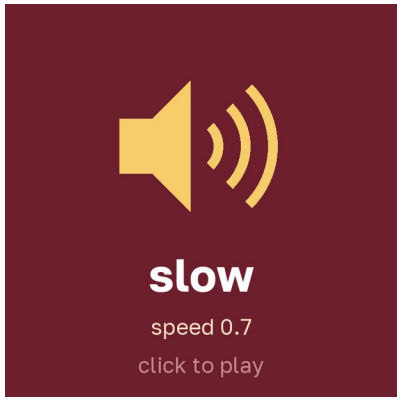
Tell the voice how to sound











"instruct": "female, child, high pitch" **"speed": 0.7** **[laughter]** in the text

Words: female, male, child, teenager, young adult, elderly, low / high pitch, whisper, accents

Describe the voice in words. The model makes it.

Is the voice good? 3 checks

1. Read it back

ASR writes what it heard: the same text?

7 of 7

2. Same person?

compare the 2 voices as vectors

close?

3. People listen

1 to 5 points, the average = MOS

1-5

NEW WORD

MOS = the average 1-5 score that people give

7 of 7 = our clips today, read back word for word

Machines check the words. People check if it sounds nice.

ACTIVITY, 3 MINUTES

Real or AI?



Real ___ AI ___



Real ___ AI ___



Real ___ AI ___



Real ___ AI ___

Good AI voices are hard to tell from real ones.

Real or AI: the answers

Voice 1: AI

OmniVoice copied a real voice from a few seconds

Voice 2: real

A real person (FLEURS recording)

Voice 3: AI

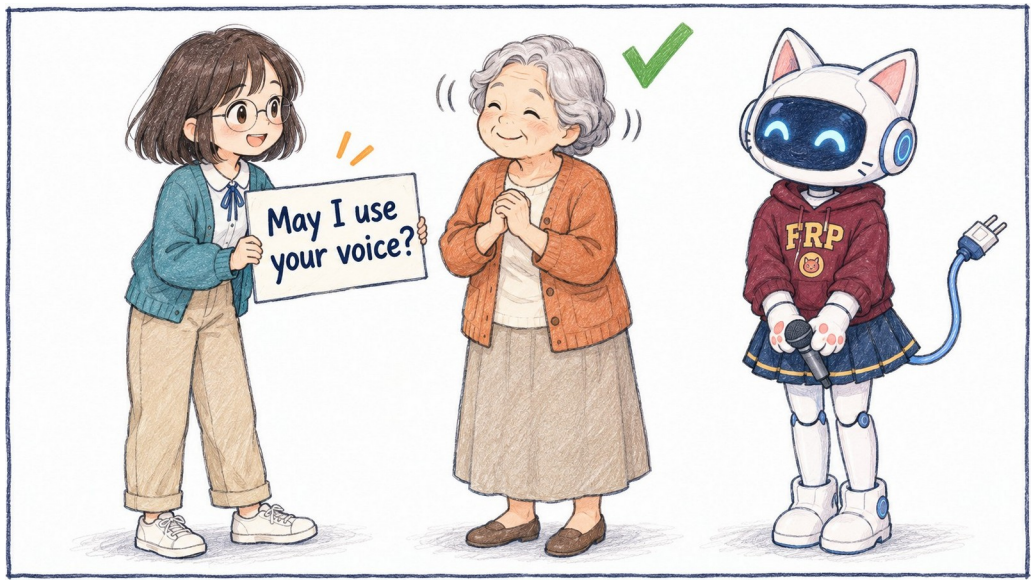
Piper, our fast voice

Voice 4: real

A real person (FLEURS recording)

A copied voice can sound real. So: ask first.

Copy a voice? Ask first.



1. **Ask the person first**
   before you copy their voice
2. **Never pretend to be a real person**
   no fake calls, no fake messages
3. **Always say: “this is an AI voice”**
   on your page and in your talk

OmniVoice: for education only (its licence).

Never copy a voice without permission. Always say it is an AI voice.

Check the voice with the ear

text → TTS → sound → ASR → text again

5 of 6

came back with no wrong letter

1 of 6

came back with a few wrong letters (Piper, English)

The ear read the robot's sentences back. Our test, 6 sentences: a quick check, not a proof.

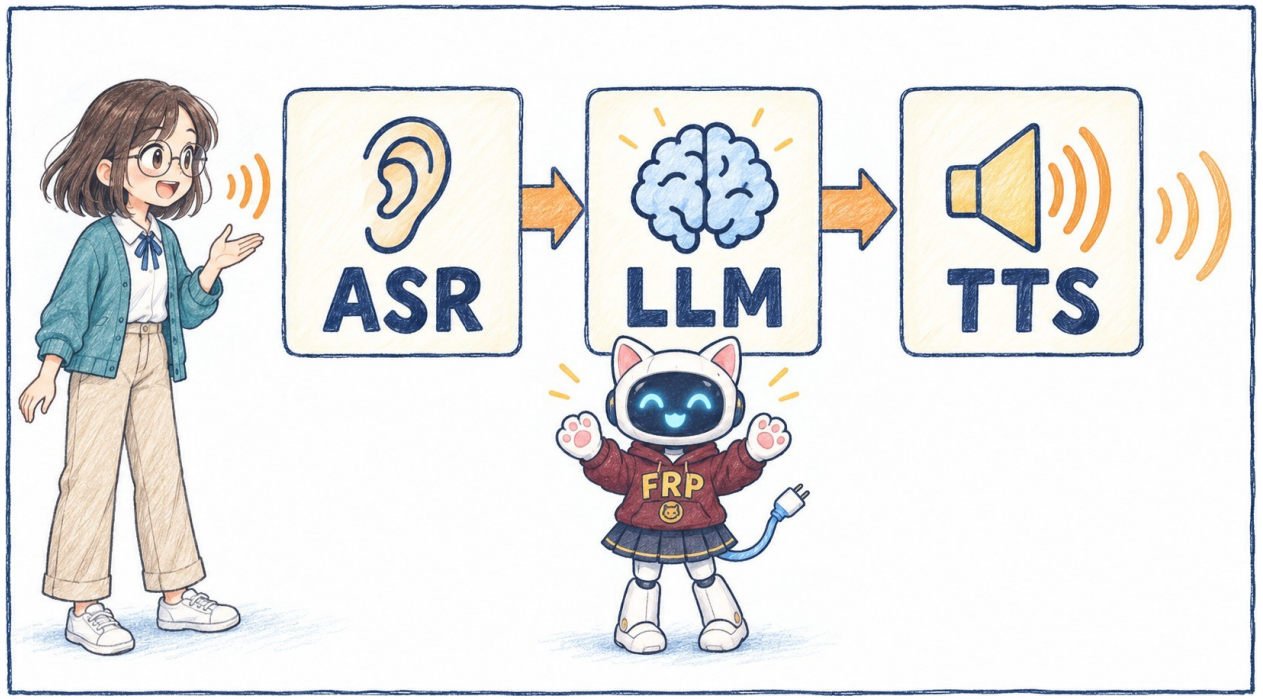
Same text back? The voice said it right.

PART E: BUILD A VOICE APP

Hear + think + speak

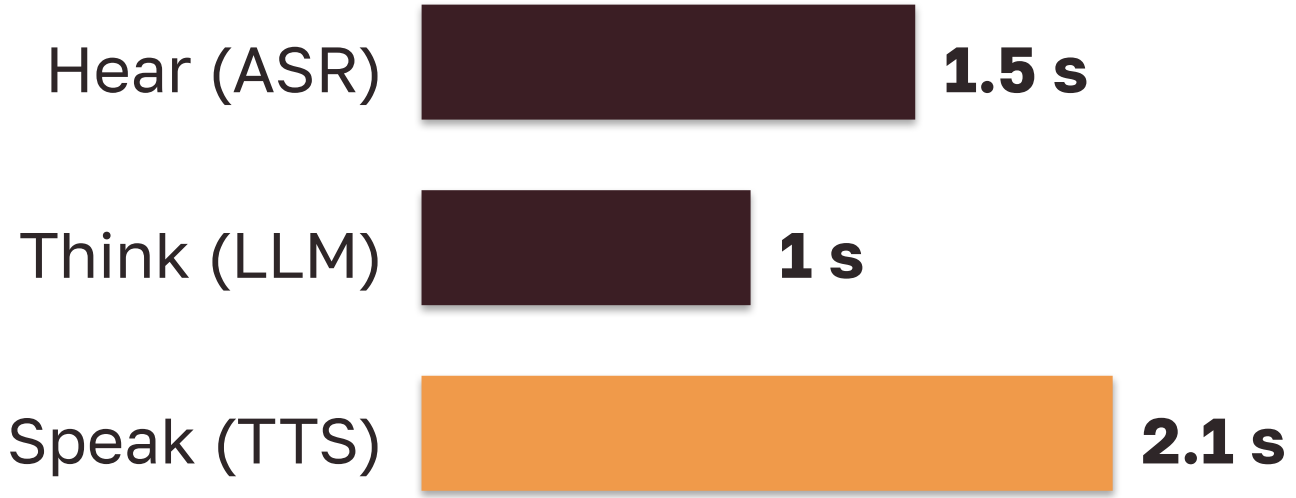
Three blocks, one app.

Your voice app = 3 blocks



ASR hears, the LLM thinks, TTS speaks.

How fast? About 5 seconds



Median of 5 runs in class workspaces, 25 Oct: 3.3 to 9.3 s in total.

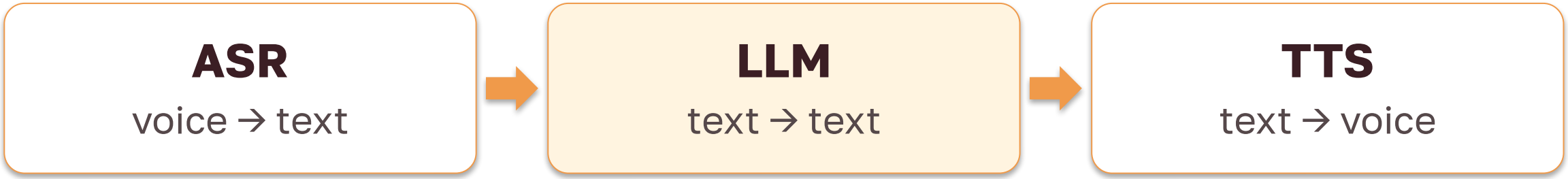
Why not instant?

Every step goes to the ISSAI GPUs in the USA and back.

The whole class shares them: busy = slower (9.3 s).

Short answers, and show “listening…” and “thinking…”.

Way 1: a chain (you build this today)



Good

Simple, any LLM

You can see and check the text

Bad

3 steps: slower (ours: about 5 s)

The text loses HOW you said it

Text in the middle: easy to build, easy to check.

ACTIVITY, 1 MINUTE

What does the text lose?

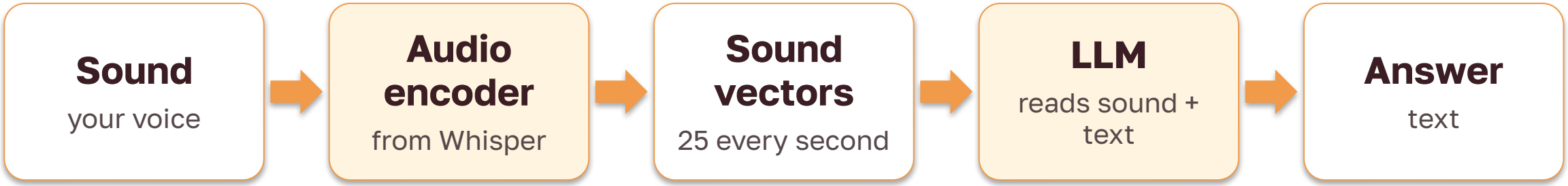
“Жақсы”	“Жақсы”	“Жақсы”	→	жақсы
happy	angry	bored		the same text 3 times

Lost: tone, feeling, who speaks, laughing, noise around you

OpenAI about its old chain: “it can't directly observe tone, multiple speakers, or background noises”

Text keeps the words and loses the voice.

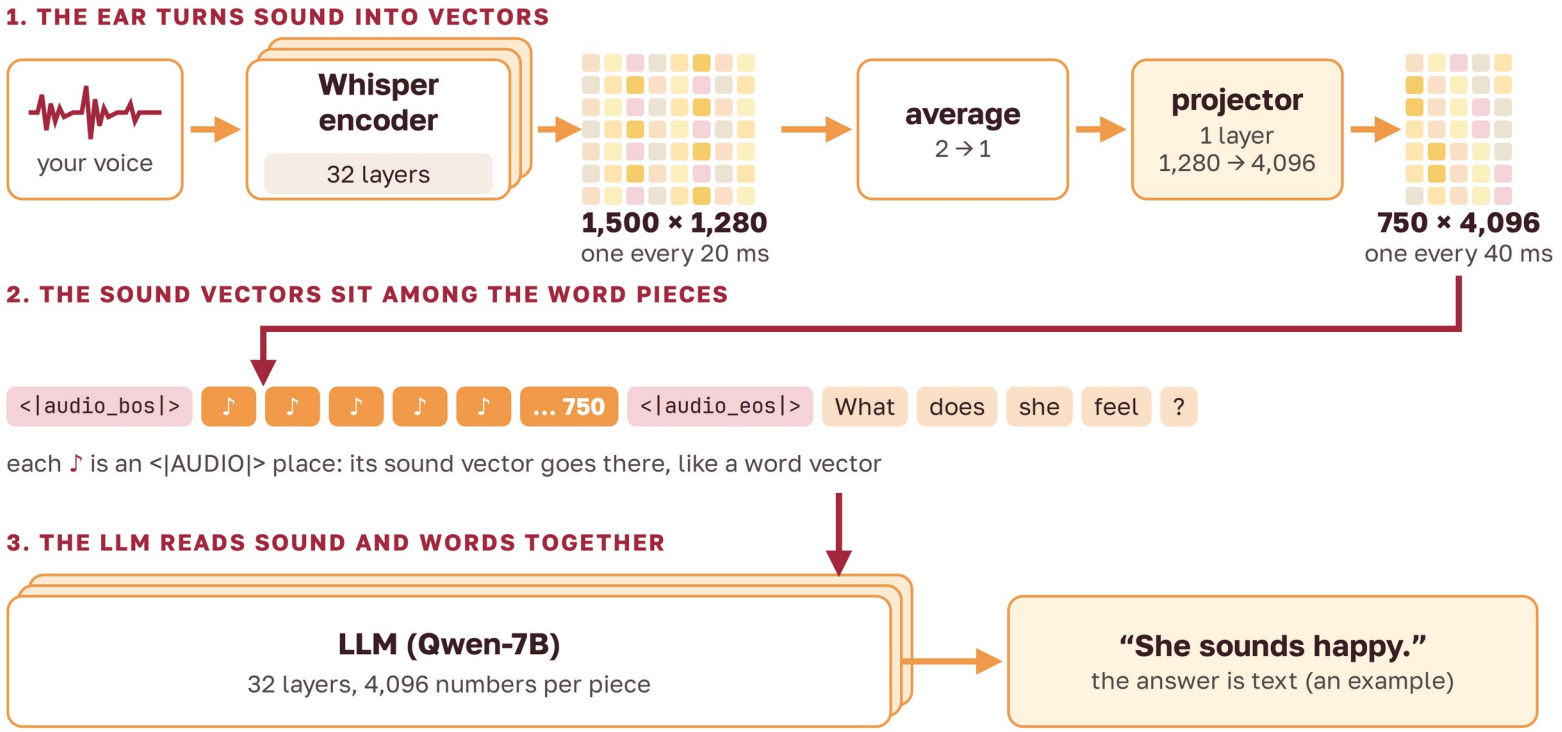
Way 2: an LLM with ears



Example: Qwen2-Audio = Whisper's encoder + the Qwen-7B LLM, **8.2 billion** parameters

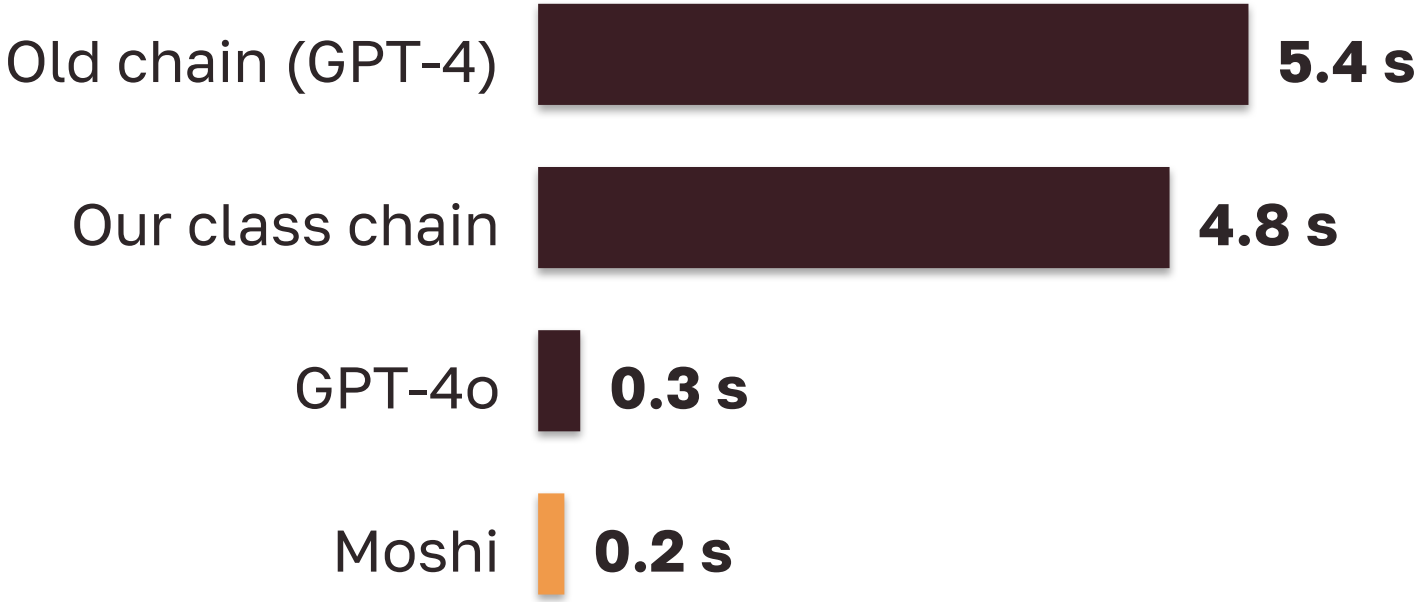
> **The LLM reads sound vectors like words.**

How the ear joins the LLM, inside



One small layer makes sound vectors look like word vectors.

Way 3: speech in, speech out



One model

hears sound tokens, writes sound tokens

can laugh, sing, and hear your tone

Time to answer, in seconds

One model, no text in the middle: as fast as a person.

Which way for your project?

	1 Chain	2 LLM with ears	3 Speech in, out
Hears your tone	no	yes	yes
Text you can check	**yes**	the answer only	no
Time to answer	about 5 s	-	0.2-0.3 s
Build it this week	**yes**	research	research

Your project: the chain. You can check every step.

CODE

Your agent needs only 3 calls

1 Hear: voice → text

Your agent reads SPEECH-API.md: no need to remember.

```
student@frp-stu-s91:~/work/day3$ curl -s $API/audio/transcriptions -F file=@question.wav \
>   -F model=gigaam -F language=kk | grep -o '"text":"[^"]*"'
"text":"қазақстанның астанасы қай қала"
```

2 Think: text → answer

```
student@frp-stu-s91:~/work/day3$ curl -s $API/chat/completions --json @ask.json -o a.json
student@frp-stu-s91:~/work/day3$ python3 -m json.tool --no-ensure-ascii a.json | grep content
                "content": "Қазақстанның астанасы Астана қаласы.",
```

3 Speak: answer → voice

```
student@frp-stu-s91:~/work/day3$ cat speak.json; echo
{"input": "Астана қаласы.", "language": "kk"}
student@frp-stu-s91:~/work/day3$ curl -s $API/audio/speech --json @speak.json -o answer.wav
student@frp-stu-s91:~/work/day3$ ls -lh answer.wav
-rw-r--r-- 1 student student 52K Oct 16 06:36 answer.wav
```

Files to text: one command

```
$ frp-asr question.wav --lang kk
қазақстанның астанасы қай қала
→ question.txt .srt .json  (3 s audio, 1 s)

$ frp-asr question.wav --lang kk --speakers --words
[00:00] A: қазақстанның астанасы қай қала
  қазақстанның 0.82-1.41  астанасы 1.49-1.93

$ frp-asr lecture.mp4 --lang ru     # 25 min
  part 1 of 3 ... part 3 of 3       # joined
```

You get 3 files

.txt the text

.srt subtitles for a video

.json times, speakers, words

Any file

voice message, interview, song, video

--lang kk = GigaAM

Voice message, song or video: one command, and you have the text.

Our speech services

	Call	Options
Hear	/audio/transcriptions	gigaam / whisper, language, word times
Speak	/audio/speech	piper or omnivoice, speed, instruct
Copy a voice	/tts + ref_audio	a few seconds of a voice (ask first!)
Files	frp-asr file --lang kk	.txt, .srt, .json
All 3	python3 voice_bot.py q.wav	hear → think → speak

Address: http://127.0.0.1:19040/v1 **Guide:** materials/SPEECH-API.md

> **Or tell your agent: “read SPEECH-API.md”.**

Speak it, hear it back: 4 commands

```
$ curl -s http://127.0.0.1:19040/v1/audio/speech -o hello.wav \
    -H "Content-Type: application/json" \
    -d '{"model": "omnivoice", "input": "Сәлем, достар!", "language": "kk",
         "instruct": "female, child, high pitch"}'
HTTP 200, 80684 bytes, 3.3 s
$ curl -s http://127.0.0.1:19040/v1/audio/transcriptions \
    -F file=@hello.wav -F model=gigaam -F language=kk
{"text":"салем достар","model":"gigaam","audio_seconds":1.68,"process_seconds":0.35}
$ curl -s http://127.0.0.1:19040/v1/audio/speech -o slow.wav -H "Content-Type: application/json" \
    -d '{"model": "piper", "input": "Hello, friends!", "language": "en", "speed": 0.8}'
HTTP 200, 60460 bytes, 2.2 s
$ curl -s http://127.0.0.1:19040/v1/audio/transcriptions -F file=@slow.wav -F model=whisper \
    -F language=en -F response_format=verbose_json -F "timestamp_granularities[]=word"
{"text":"Hello, friends.", "words":[{"word":"Hello","start":0.03,"end":0.35}, ...]}
```

Text in, voice out. Voice in, text out. One request each.

Break, 10 minutes

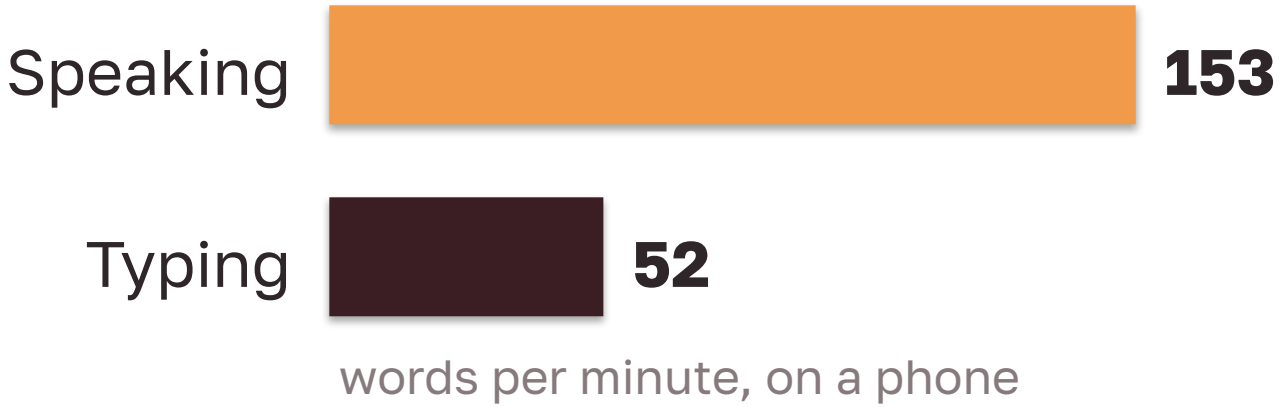
Next: talk to your computer

PART F: TALK TO YOUR COMPUTER

Voice coding and voice assistants

Your voice is the fastest keyboard.

Why speak instead of typing?



2.9×

faster in English (Stanford + Baidu, 2016)

Race, 2 minutes: one types, one speaks:

“Make the title bigger, add a dark mode button, and show my score at the top.”

Speaking is about 3 times faster than typing.

Vibe coding started with a voice

> “There's a new kind of coding I call **vibe coding** … **I just talk to Composer with SuperWhisper so I barely even touch the keyboard.** I ask for the dumbest things like ‘decrease the padding on the sidebar by half’ …”
>
> Andrej Karpathy, on X, 2 February 2025

He also says: good for **weekend projects**; for real projects you still check and measure.

Say what you want. The agent writes the code.

Voice coding tools

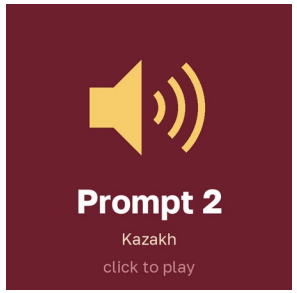
Tool	How	Russian	Kazakh
Mac dictation	Microphone key, or Fn Fn	yes	no
Windows voice typing	Windows key + H	yes	no
SuperWhisper, Wispr Flow	apps: speak into any app	yes	-
Claude Code	/voice	yes	no
Our ear (GigaAM)	frp-asr, or your Listen button	yes	**yes**

English, Russian: built-in dictation. Kazakh: our ear.

We tried it: voice prompts



Whisper: word for word



GigaAM: word for word
Whisper: **ойнады** (played), not ойнат (play!)



get_user_score → **getUse.cos**, **get use t coss**
JSON → **jason**

Our ears, 25 Oct. The prompts were spoken by our TTS voice.

Ears know words, not your variable names.

How to speak a prompt

1 **Say it in normal words** “a function that gets the score”, not get_user_score

2 **Speak long, like to a friend** what you see, what is wrong, what you want

3 **Read before Enter** 2 seconds: did it hear you right?

4 **Kazakh: our GigaAM ear** English, Russian: built-in dictation is fine

> **Speak like you explain to a friend. Then read it once.**

ACTIVITY, 5 MINUTES

Voice-code one change

Dictation on
Mac: mic key
Windows: Win + H

→

Say a change
20-30 s, like to a friend

→

Read it
fix one word

→

/build
look at your page

Example: *"On my quiz page the questions are too small and the Next button is hard to find. Make the question text bigger, put Next in the middle, and make it green."*

Who got the longest useful prompt in 30 seconds?

ChatGPT voice: a short story



In one year: from a chain of 3 models to one model that talks.

Old voice vs new voice

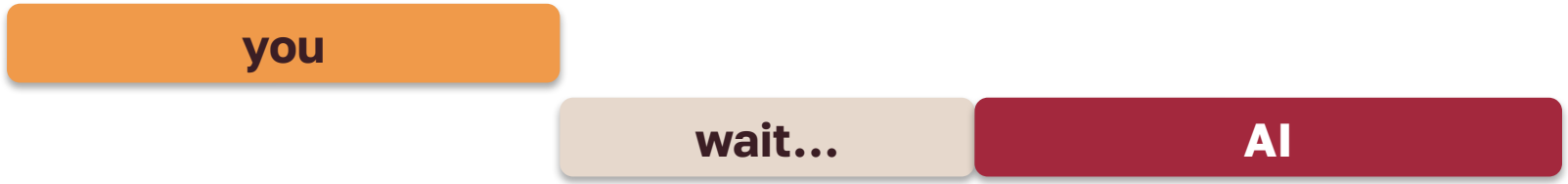
	2023: a chain	2024: one model
Inside	Whisper → GPT → TTS	sound in → sound out
Time to answer	2.8-5.4 s	**0.32 s**
Hears your tone	no: only the text	yes
Stop it anytime	-	yes
Our class voice bot	**this kind**	-

Same 3 jobs: hear, think, speak. Now in one model.

Who talks when?

1 Push to talk

our voice bot



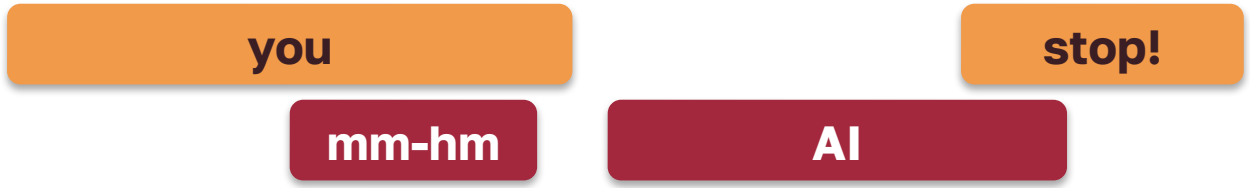
2 Waits for silence

VAD: most assistants



3 Full duplex

Moshi, GPT-4o voice



A real talk: it listens while it speaks, and you can stop it.

VAD: the small ear that listens first

No invented text

silence never reaches the big ear: no “Bye.” × 178

Less work

the big ear runs only when someone speaks

Knows you stopped

answer after about 0.5 s of silence

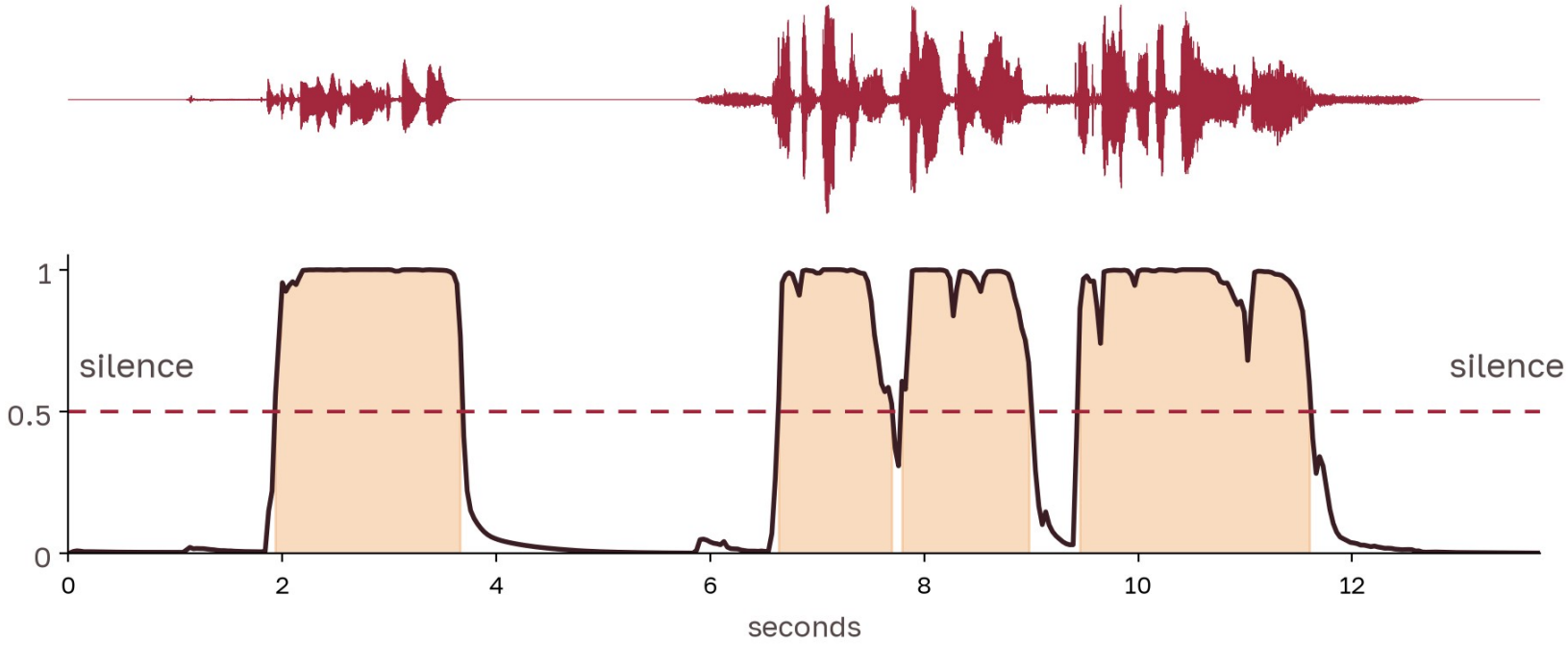
Our hiss clip: the VAD's highest speech chance was **0.11**, so nothing was sent.

NEW WORD

VAD = voice activity detection

> **First ask “is someone speaking?”, then listen.**

VAD on our own clip

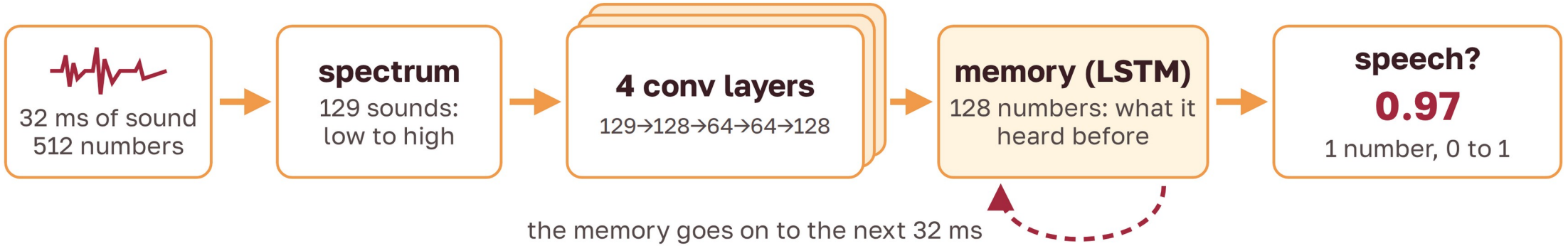


Silero VAD in a class workspace, 25 Oct: 430 steps of 32 ms, under 1 ms each.

Orange = speech. Only these parts go to the big ear.

Inside a VAD: tiny and fast

EVERY 32 ms: 512 NUMBERS IN, 1 NUMBER OUT



Silero VAD: about 0.24 million numbers to learn. Our Whisper: 809 million.

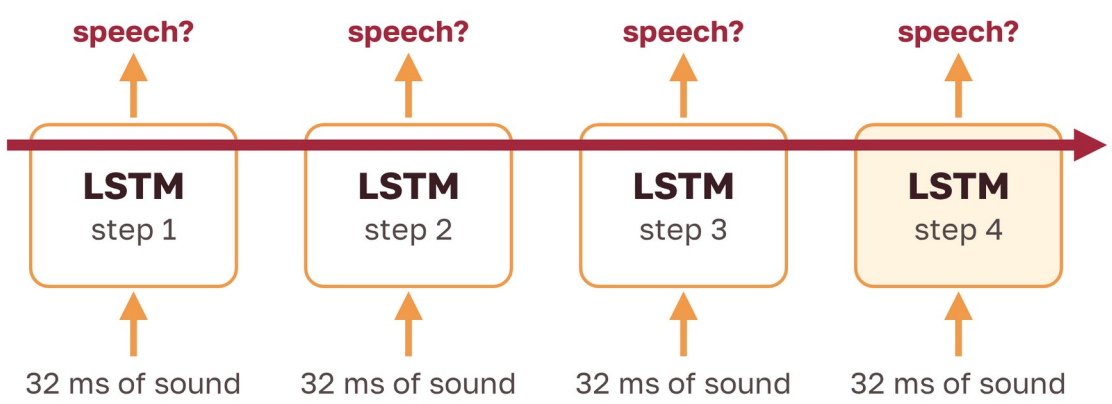
Under 1 ms of work for each 32 ms of sound, on one CPU thread. It runs all the time, the big ear only when someone speaks.

A tiny CNN + a small memory: speech or not?

What is an LSTM?

IT READS ONE STEP AT A TIME, AND CARRIES A MEMORY

red line = the memory (128 numbers), from step to step



ONE STEP, INSIDE: 3 GATES

1. forget
what to drop from the memory

2. write
what new to keep from these 32 ms

3. show
what to send out now: speech or not

A gate is a set of numbers from 0 (closed) to 1 (open).

NEW WORD

LSTM = step by step, with a memory

Transformer: all pieces at once. **LSTM:** step by step: good for live sound.

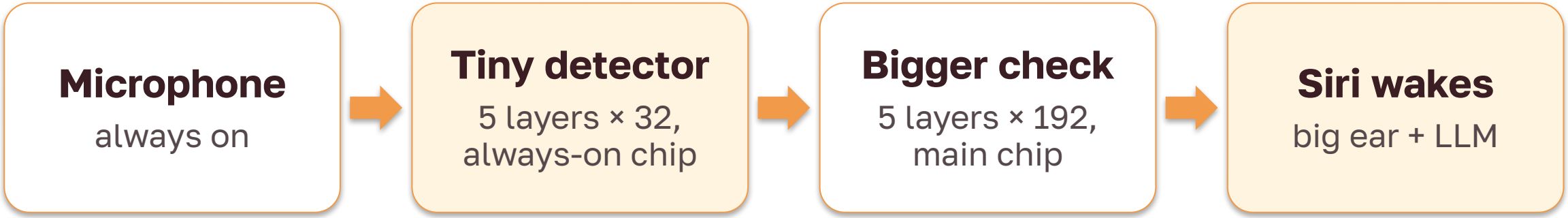
An LSTM reads step by step and keeps a small memory.

VAD models you can use

Model	Inside	Good to know
Silero VAD	spectrum + CNN + LSTM	2 MB, free, 6,000+ languages
WebRTC VAD	classic math (GMM)	10-30 ms frames, mode 0-3
pyannote	neural networks	also: who speaks when
OpenAI Realtime	server_vad / semantic_vad	waits 500 ms / reads words

Small, fast and free: put a VAD before every big ear.

“Hey Siri”: a tiny ear that waits



Why tiny? It runs all day: the battery must last.

Privacy: before the wake word, the sound stays on the phone.

A tiny ear waits for one phrase. Then the big one wakes up.

Make your own “Hey Aru”

Hey Aru!	Hey Aru!	Hey Aru!	Hey Aru!	Hey Aru!	Hey Aru!
child	old man	whisper	Russian accent	British accent	Piper
click to play	click to play	click to play	click to play	click to play	click to play

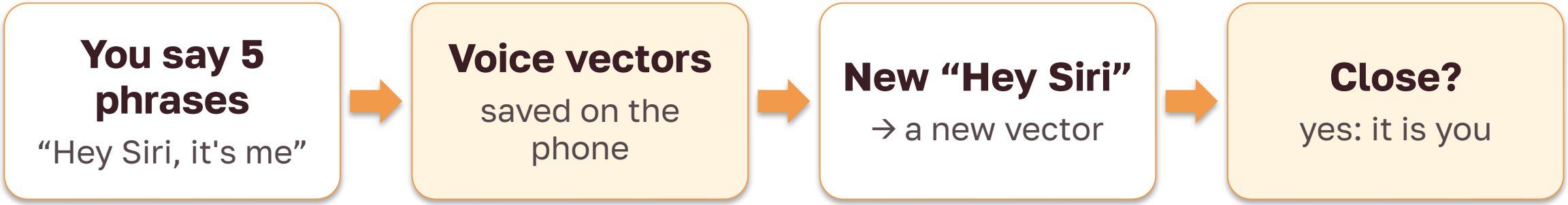
1 **TTS says “Hey Aru” in 1,000s of voices** (above: 6 from our TTS today)

2 **Add noise, train a tiny model** openWakeWord: under 1 hour in Colab

3 **It runs on a Raspberry Pi** 15-20 wake words at once

Wake words can learn from AI voices only.

Only my voice



A voice becomes a vector. Same person = close vectors. Next part: how this works.

> **Your voice is a key: compare vectors, not words.**

LIVE, 5 MINUTES

Talk to a voice assistant

1. Switch language

start in Kazakh, go on in Russian

2. Stop it

talk in the middle of its answer

3. Change the voice

whisper, then a sports reporter

4. Feelings

ask it to laugh, or to sound sad

After each one: what was fast? What was strange?

It hears your tone: the sound itself goes in.

Your voice is personal

Don't say secrets

no passwords or addresses to an assistant

Ask first

before you record other people

It can be copied

a few seconds are enough

In your project

say when you record; delete what you don't need

Your voice is data about you. Share it carefully.

HANDS UP, 3 MINUTES

Quiz time

- Why do people code with their voice?
- How do you say a variable name in a voice prompt?
- How does an assistant know you stopped talking?
- **Old ChatGPT voice vs new: what changed inside?**

PART G: SOUND AS VECTORS

Audio embeddings

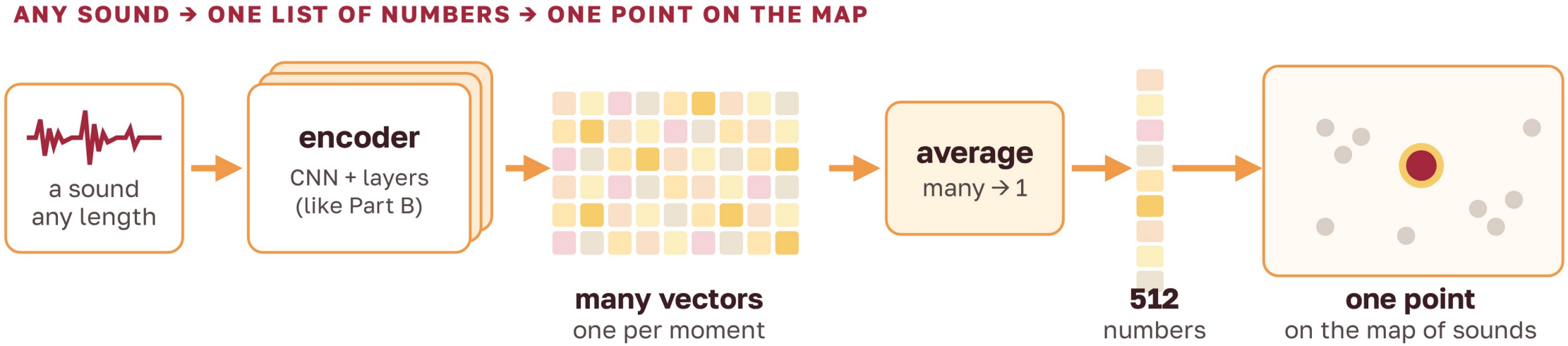
Every sound becomes a point on a map. Similar sounds sit close.

Every sound = a point on a map



Similar sounds sit close together.

How a sound becomes one point

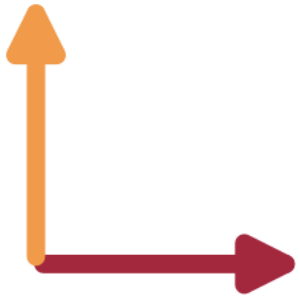


Encoder → average → 512 numbers → one point.

Close = similar



very similar: 0.95



not related: 0



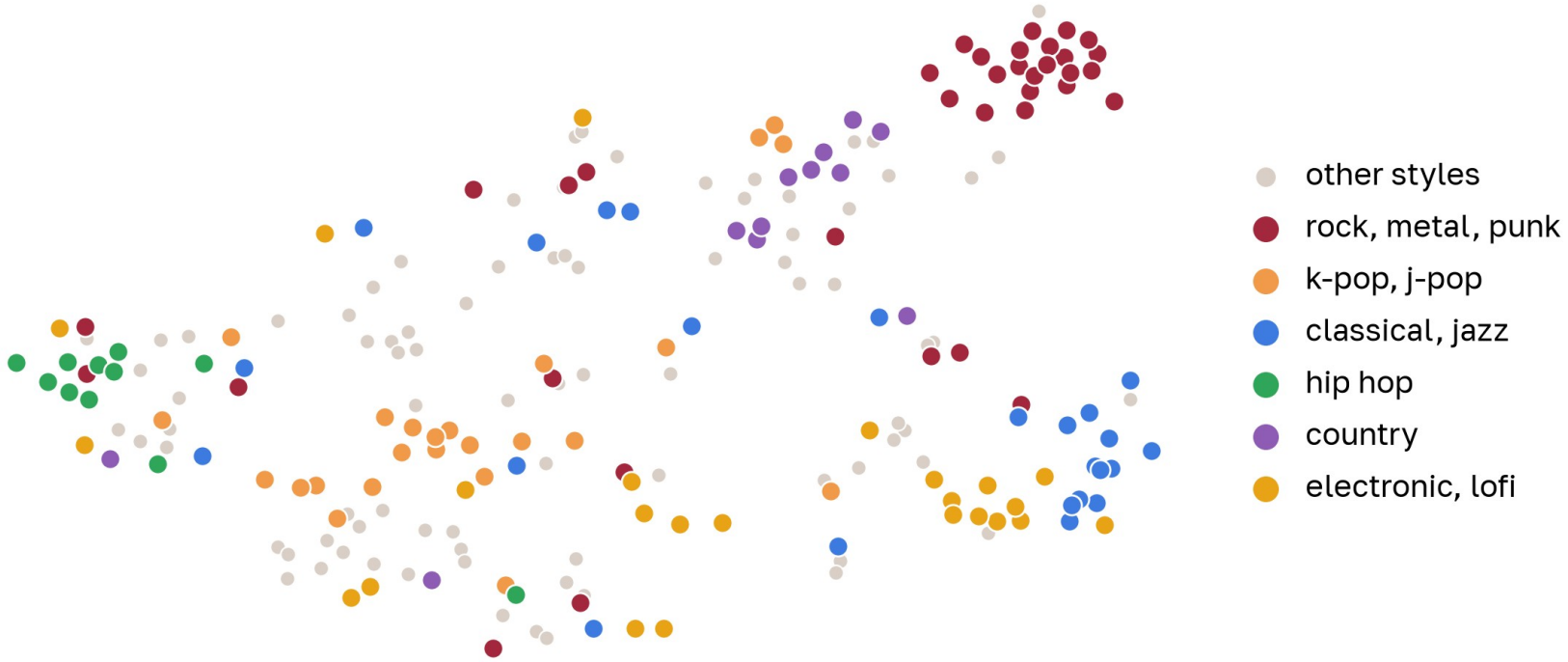
opposite: −1

NEW WORD

cosine similarity = how close 2 arrows point: 1 = same

Small angle = similar sounds.

Our 215 songs on the map



41 %

nearest song = same style

5 %

by chance

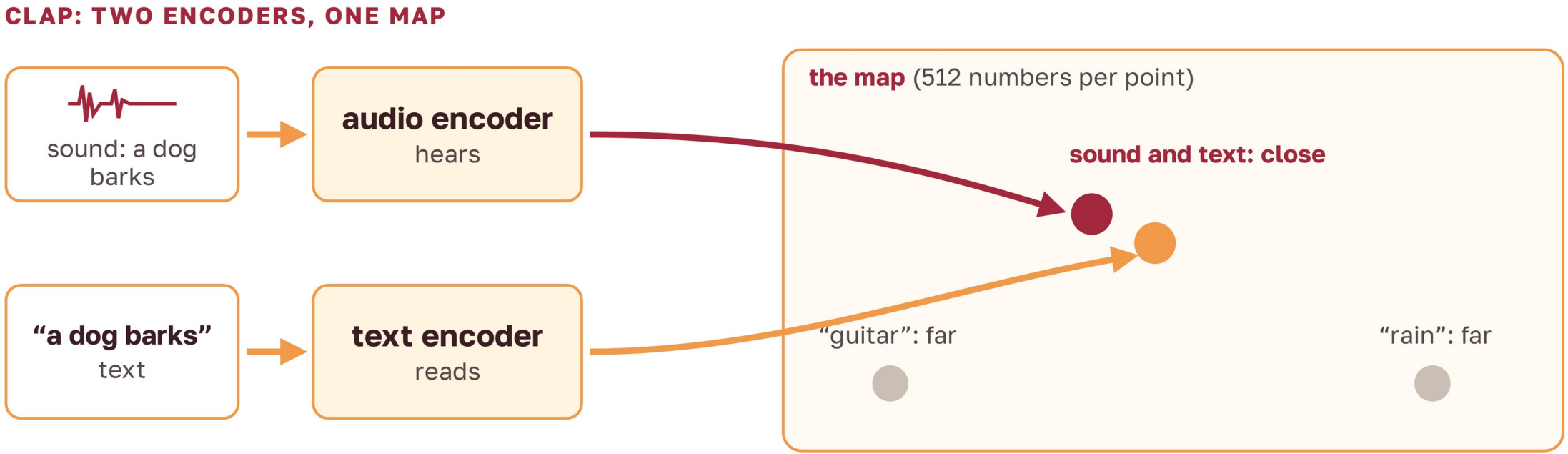
Only sound, no tags: the styles find each other.

Why are embeddings so important?



Find, search, recommend, check: all with “close points”.

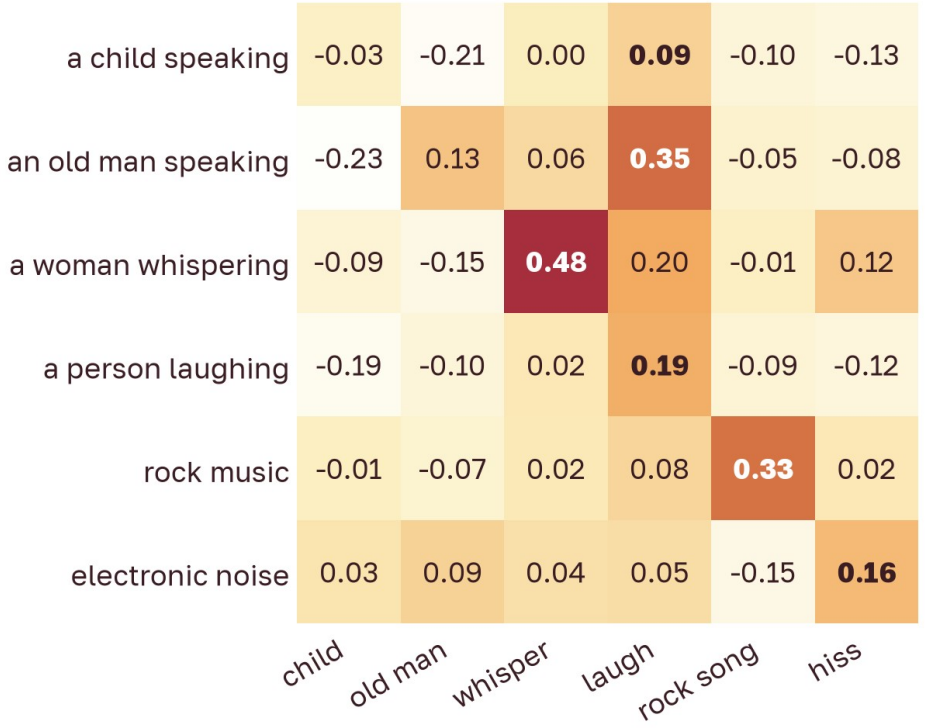
Text and sound on one map: CLAP



Trained on 633,526 sounds with their text: matching pairs pulled together.

Two encoders, one map: search sounds with words.

We tried it: 4 of 6 right

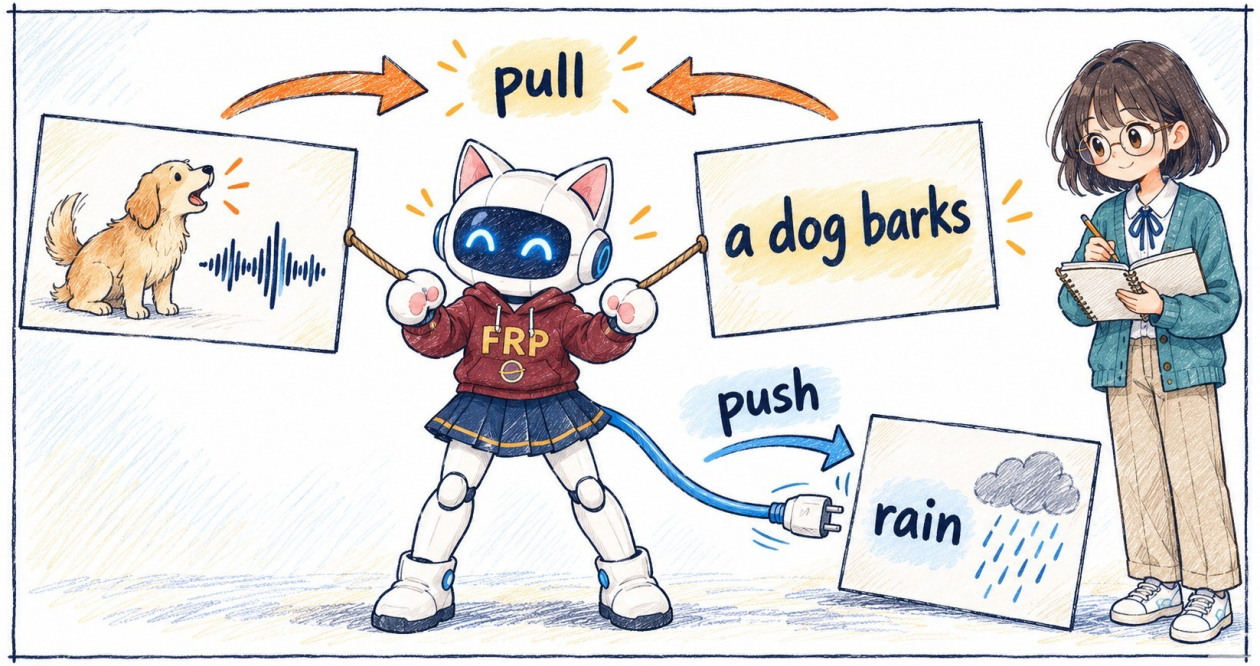


✓ whisper, laugh, rock, noise

✗ child, old man

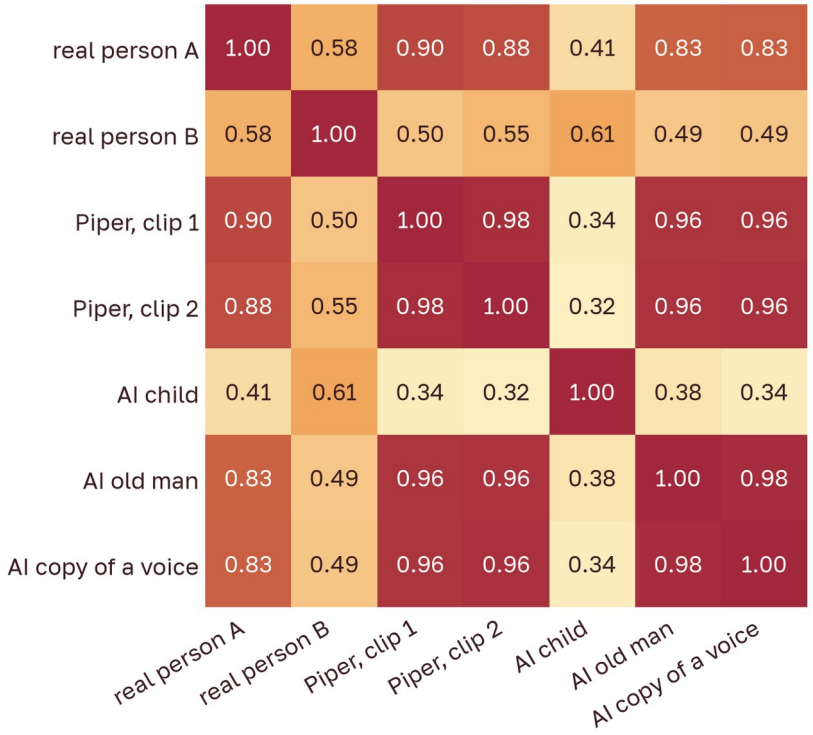
It knows kinds of sound, not who speaks.

How do they learn? Pull and push



Pull matching pairs together, push the others apart.

Who is speaking? Voice vectors



0.98 same voice

0.58 2 different people

0.96 2 different men: too close!

Same voice = close. Test your model on your voices.

Sound vectors in your project

Find similar	Search with words	Who spoke?	Check a copy
songs, sounds	“laughing”	group by voice	close to the real voice?

Copy: /plan Read materials/GPU-JOBS.md. Write a GPU job that turns my clips in data/ into CLAP vectors (laion/clap-htsat-unfused) and prints, for each clip, the 3 most similar clips. Then test it on 5 clips I know.

One vector per sound. Then compare points.

HANDS UP, 3 MINUTES

Quiz time

- What is an audio embedding?
- Which model can search sounds with words?
- Which model knows WHO is speaking?
- Which small model stops “Bye.” × 178?
- **How can we make “Hey Aru” with no recordings?**

WORKSHOP

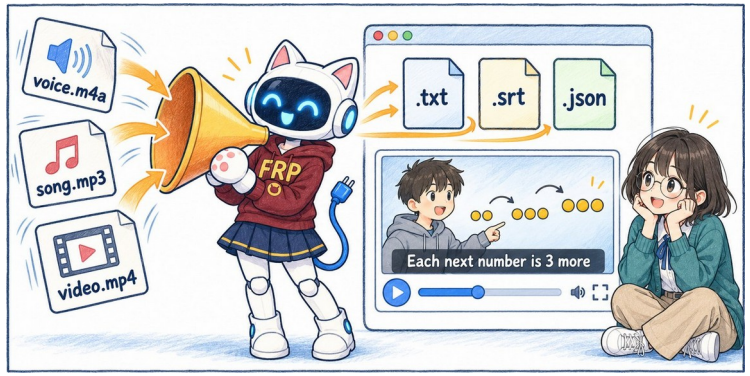
Speech in your project

Choose: your page speaks, listens, or turns files into text.

What you can build today







A Your page speaks

text → voice

B Your page listens

voice → text

C Files to text

audio, video → text, subtitles

Pick the one that fits your project.

WORKSHOP, 15 + 2 MINUTES

Add speech to your project

A Your page speaks

a button reads text aloud

Copy: /plan In my project page: add a Speak button that reads the text aloud with the class TTS. Read materials/SPEECH-API.md.

B Your page listens

your voice → text on the page

Copy: /plan In my project page: add a Listen button: record my voice, send it to the class ASR, show the text. Read materials/SPEECH-API.md.

C Files to text

audio, video → text, subtitles

Copy: /plan Use frp-asr to turn my audio files in data/ into text and subtitles, then use the text in my project.

Then: /build → /check (press it, listen) → /save prompts to copy: materials/day3/WORKSHOP.md

Done: your project speaks or listens, and you saved it.

Speech in your project



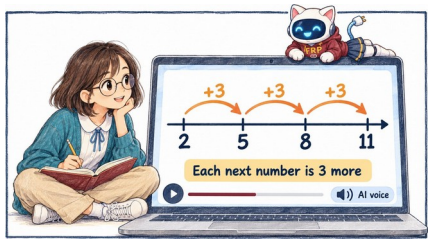
Kids learning game

questions read aloud



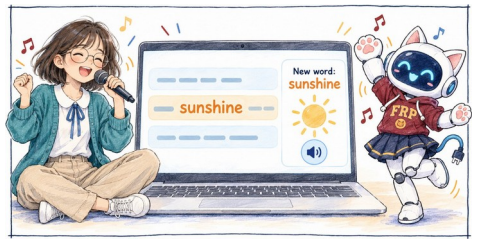
Presentation coach

words per minute, filler words



School topic videos

AI voice + subtitles



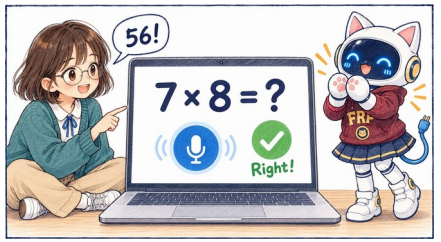
Songs to words

karaoke word times



Scam detector

voice message → text → check



Quiz against brainrot

answer by voice

Add one line to your plan: “Voice: …”

Measure your speech part

WER = wrong words ÷ all words

1. 10 clips + the true text (you type it)
2. frp-asr on all 10 clips
3. Agent: “WER per clip with jiwer + average”
4. Compare: GigaAM / Whisper, music / no music

Your Demo Day number

“On 10 Kazakh voice messages, GigaAM made 6 % word errors, Whisper 31 %.”

(an example: measure your own)

A number makes your speech part real.

5 THINGS TO REMEMBER

1. Sound is numbers. ASR turns them into text, TTS turns text into sound.
2. Inside, the ear is like an LLM: the encoder reads, the decoder writes.
3. Hearing is guessing: VAD first, the right ear, then check the text.
4. Sounds become vectors: similar sounds, close vectors.
5. Ask before you copy a voice. Always say it is an AI voice.

Sources

Measured by us on the ISSAI FRP server (FRP Speech Lab), 8 and 13 Oct 2026: word error rates, the ocelot sentence, the voice bot pipeline.

Measured by us in class workspaces, 25 Oct 2026: voice bot speed (5 runs, DeepSeek + GigaAM + Piper), frp-asr runs.

Radford et al., Robust Speech Recognition via Large-Scale Weak Supervision (Whisper), arXiv 2212.04356, 2022.

Koenecke et al., Careless Whisper: Speech-to-Text Hallucination Harms, arXiv 2402.08021, 2024.

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